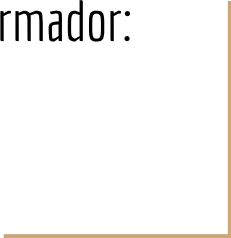


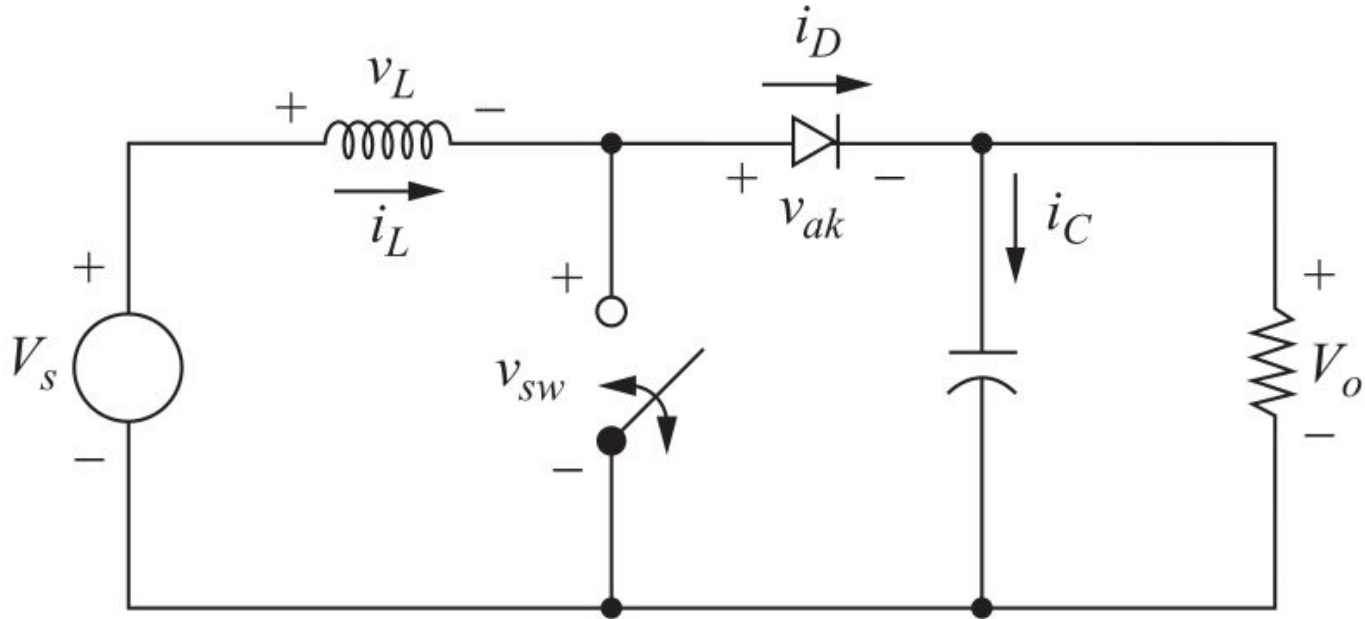


Conversión CC/CC

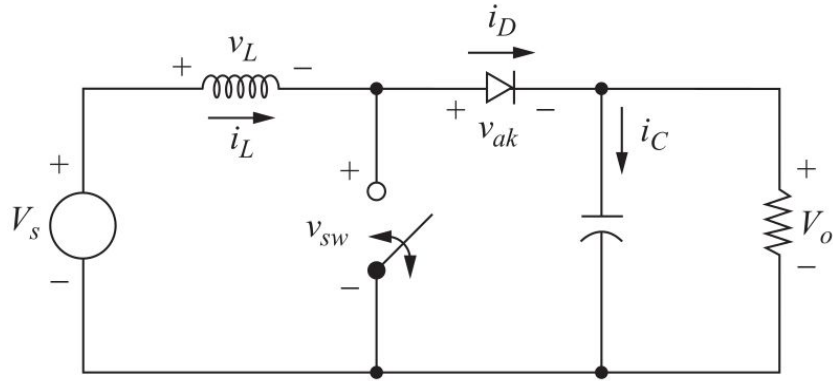
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Elevador



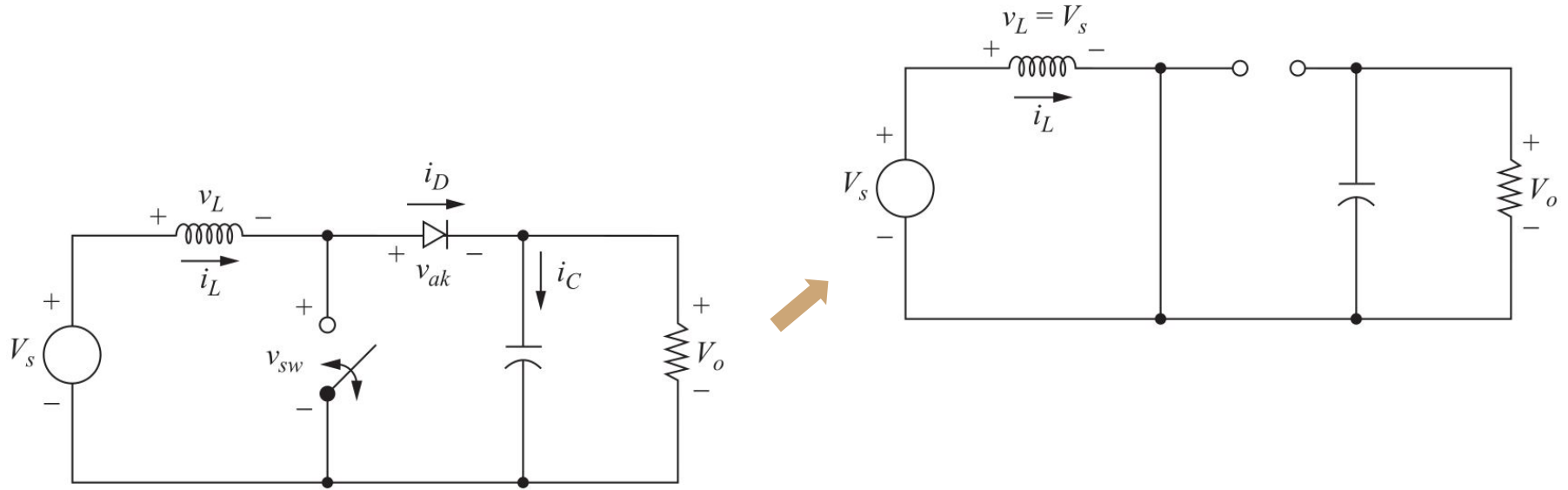
Convertidor Elevador: esquema circuital



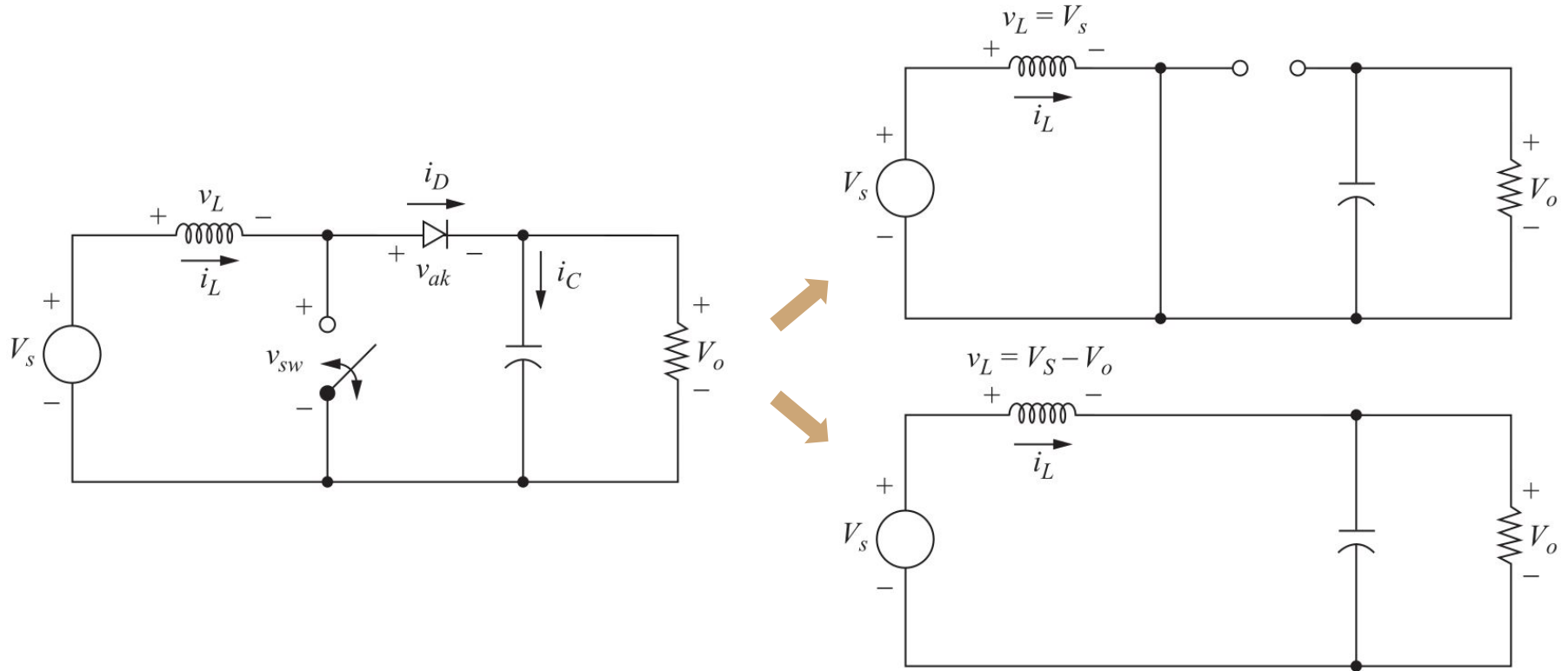
Convertidor Elevador: funcionamiento



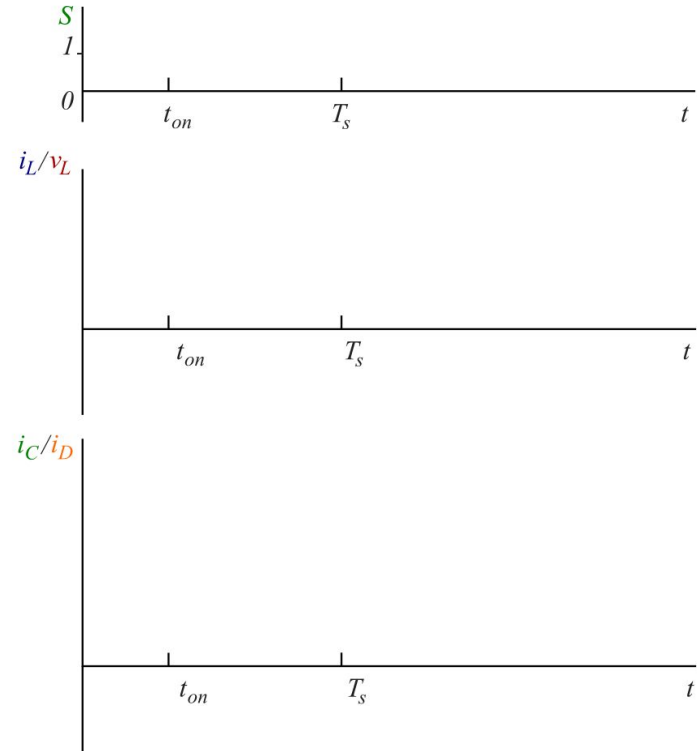
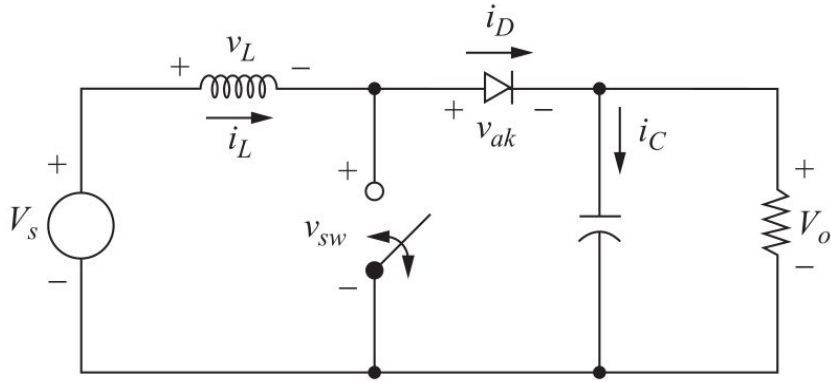
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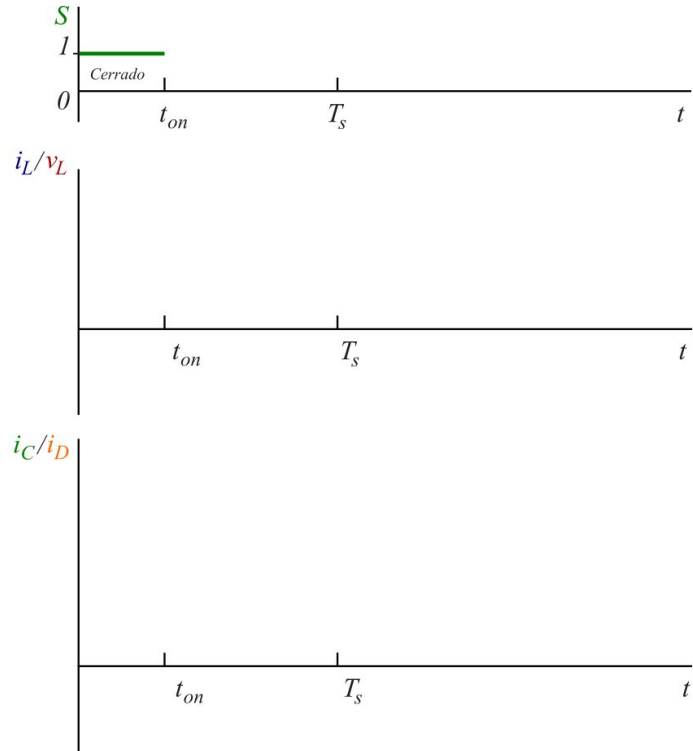
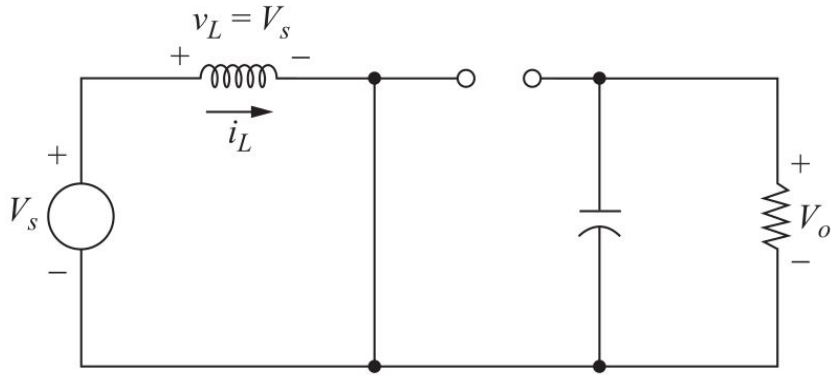
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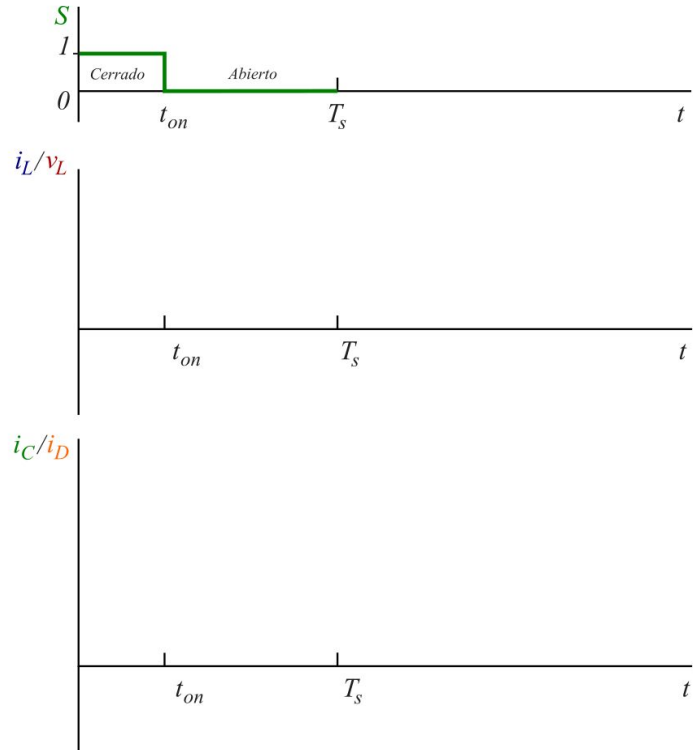
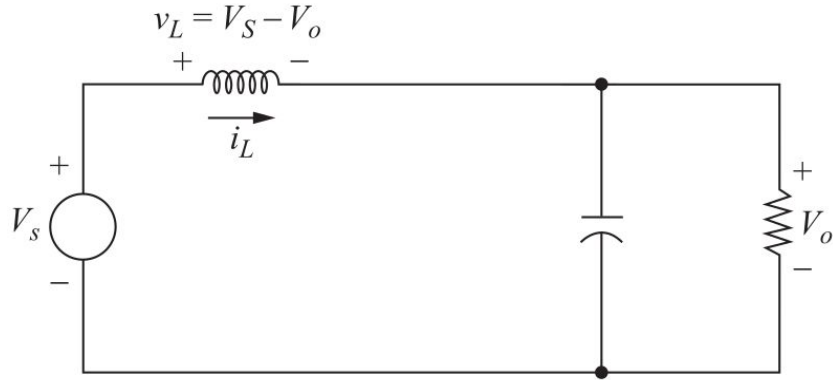
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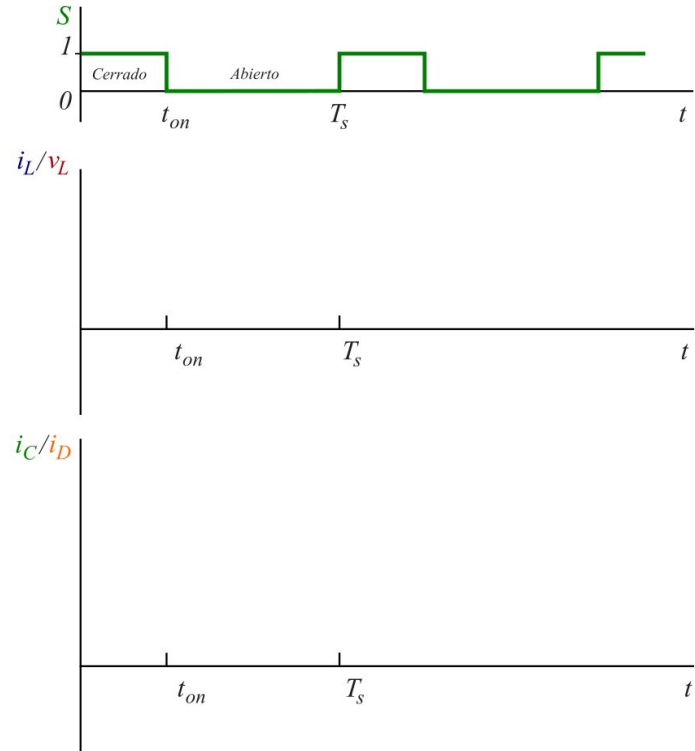
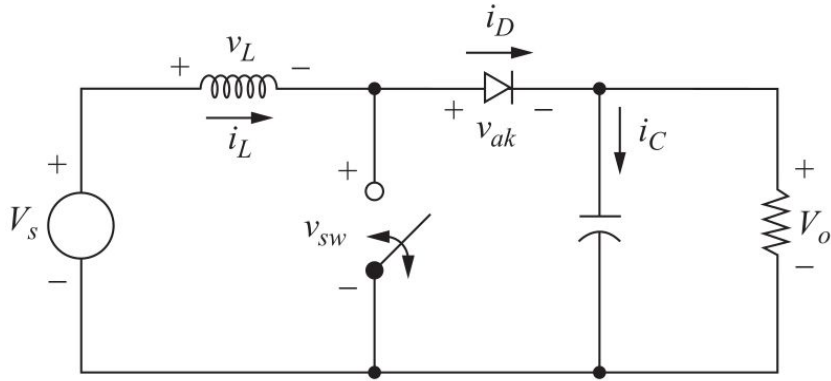
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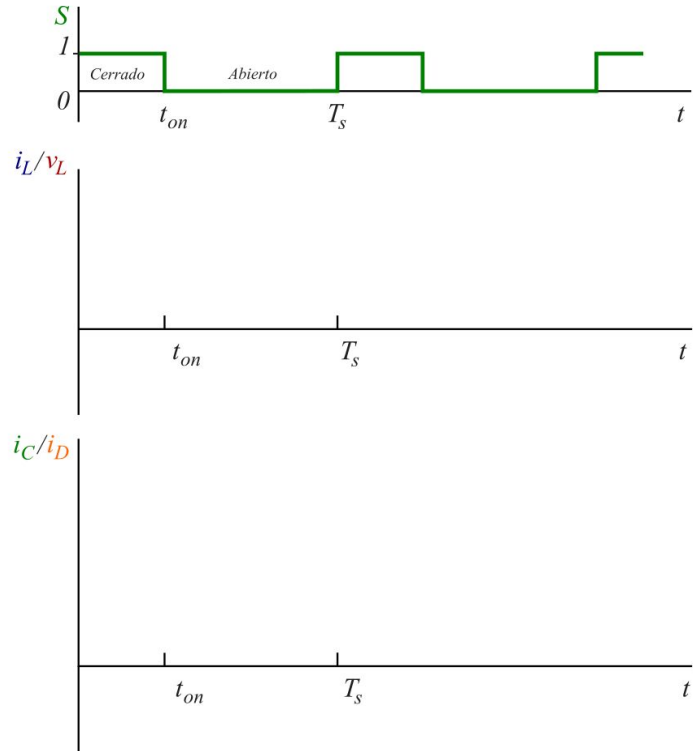
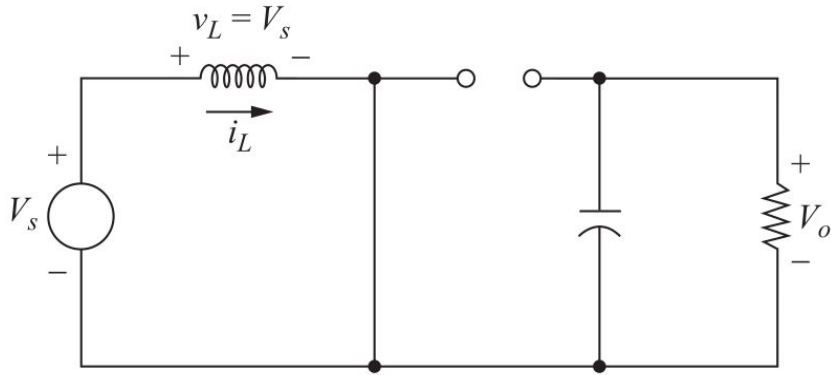
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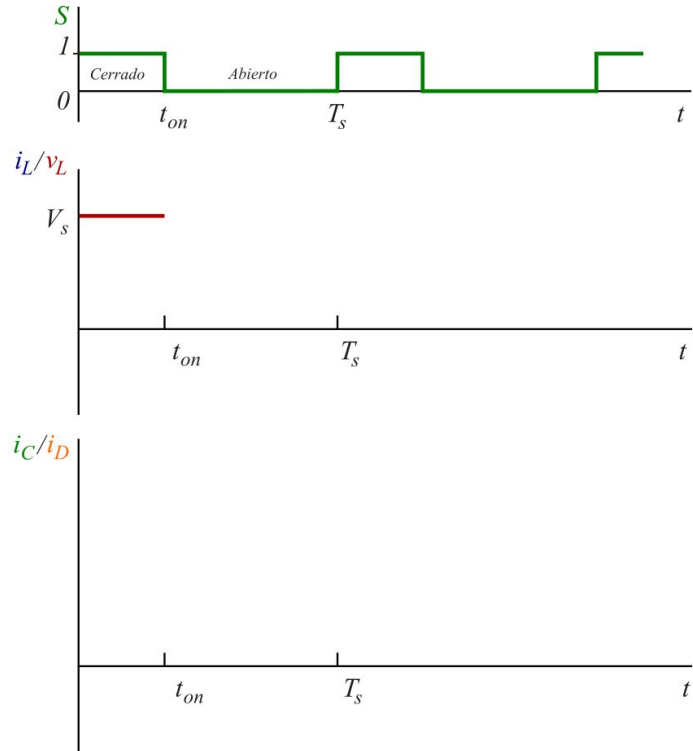
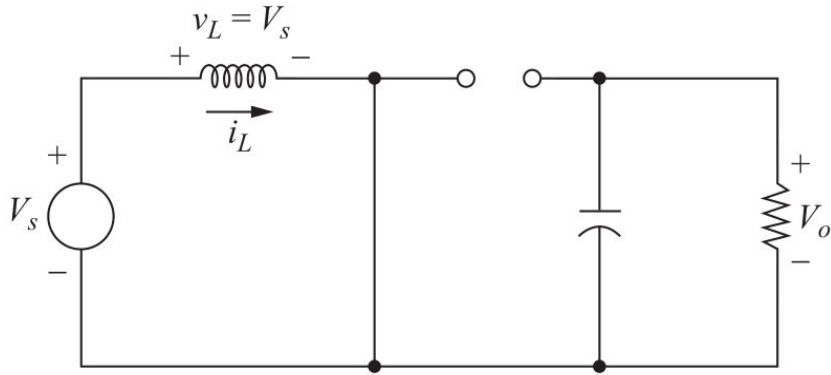
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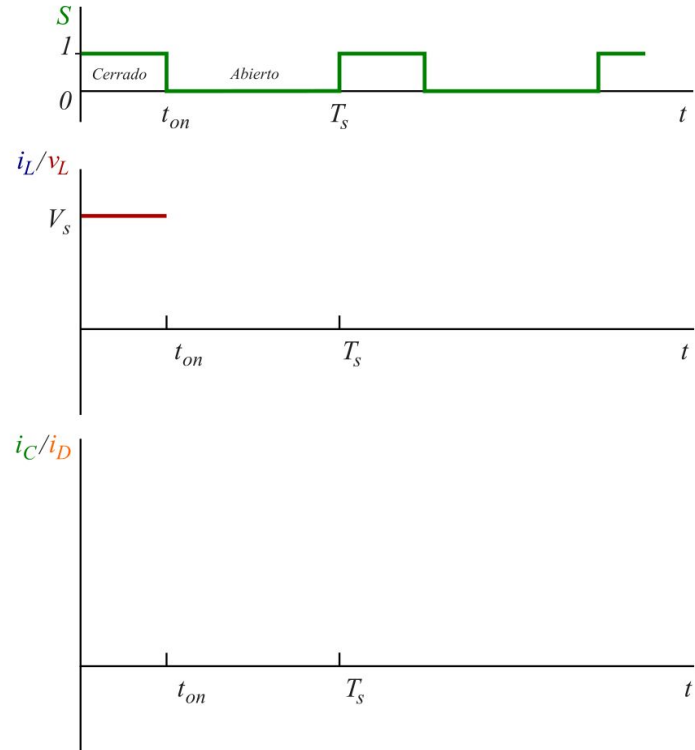
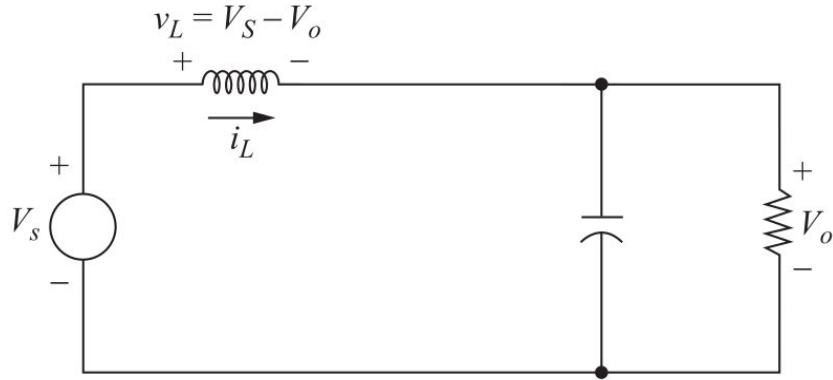
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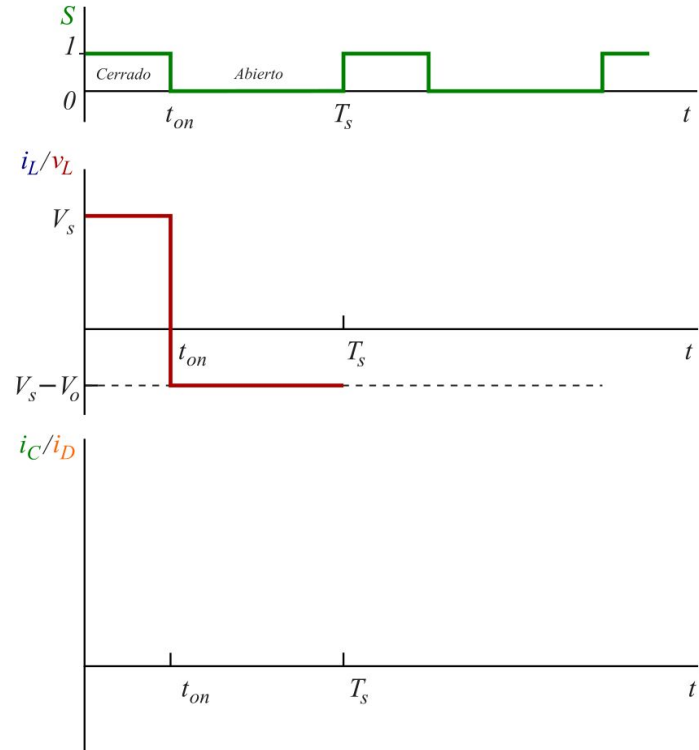
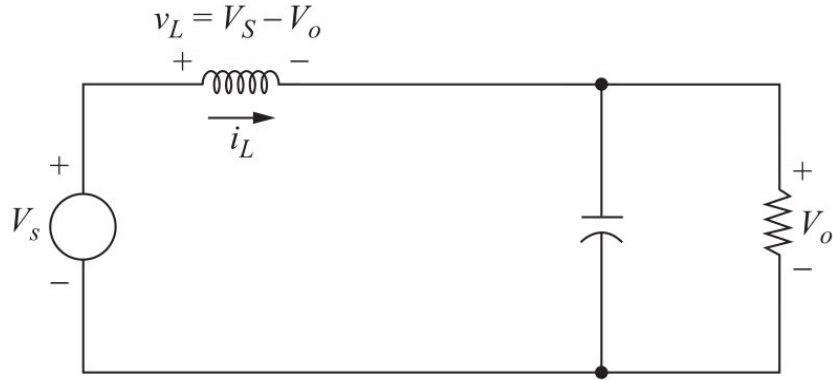
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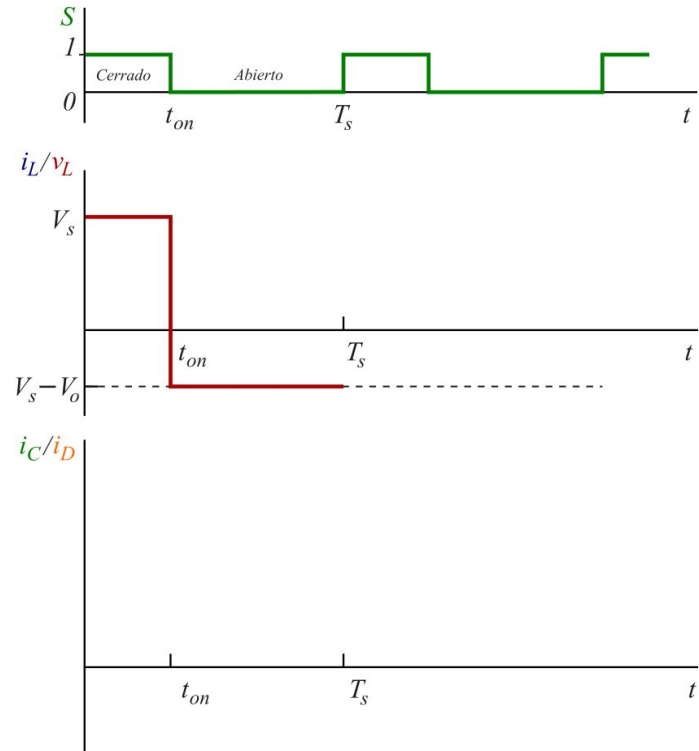
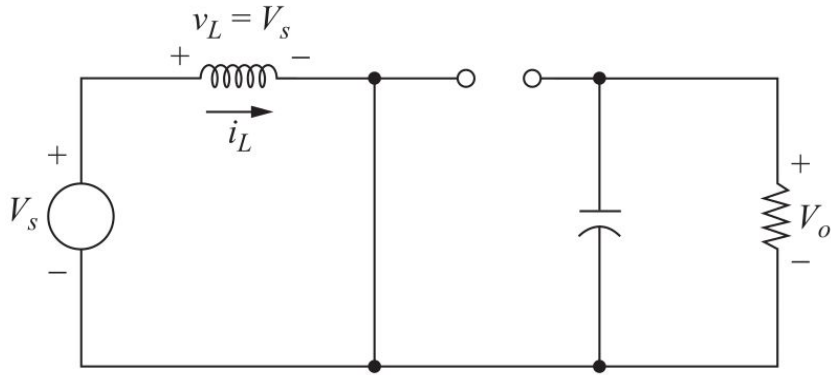
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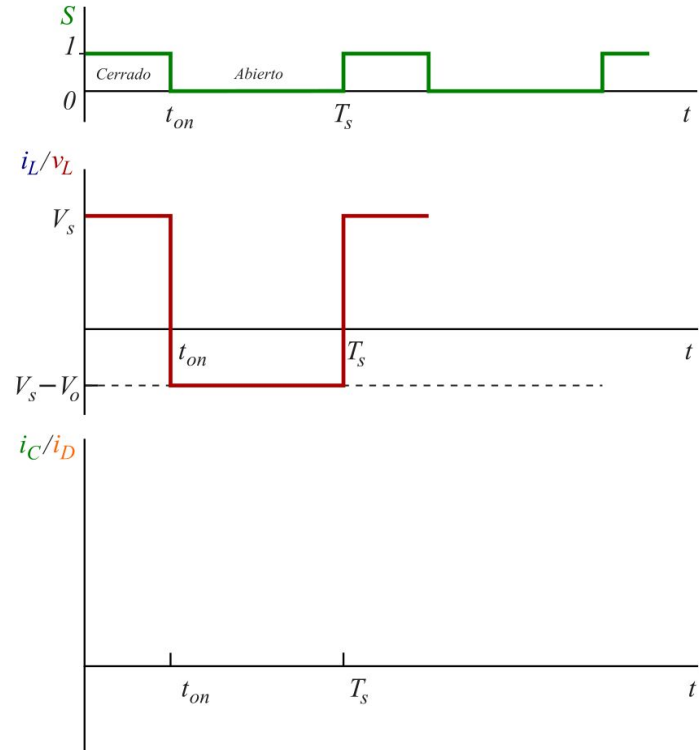
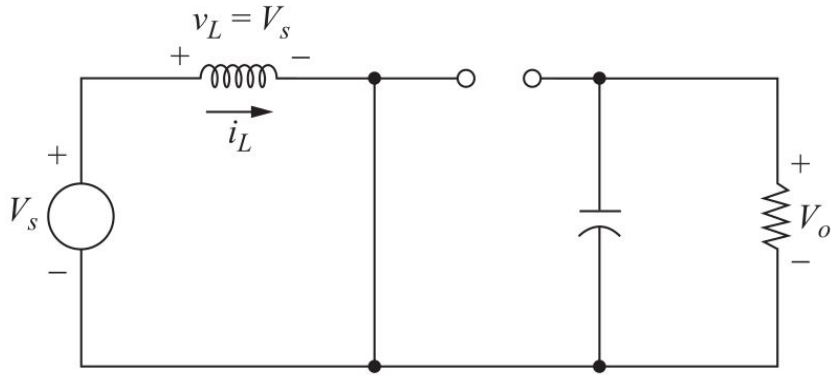
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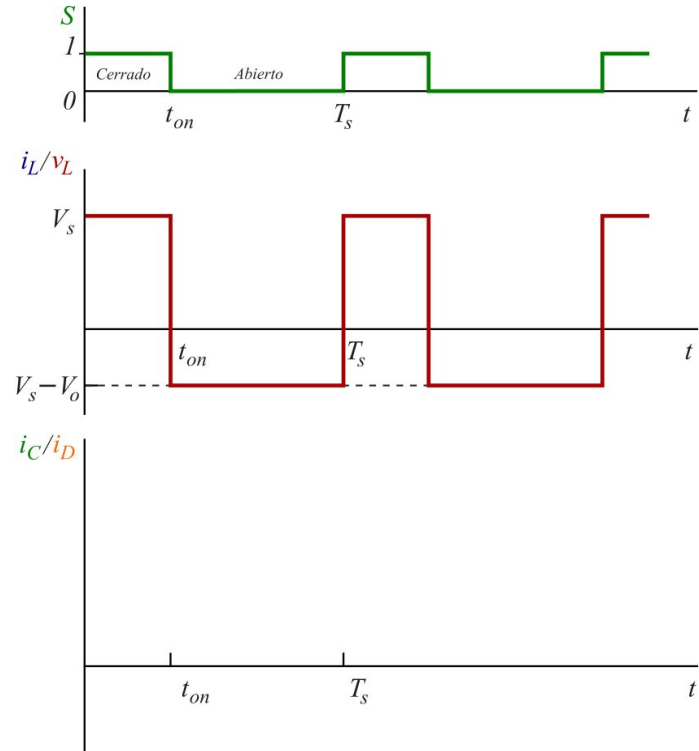
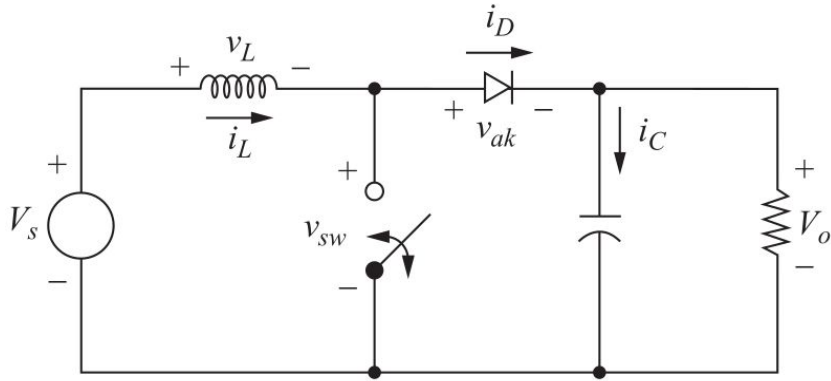
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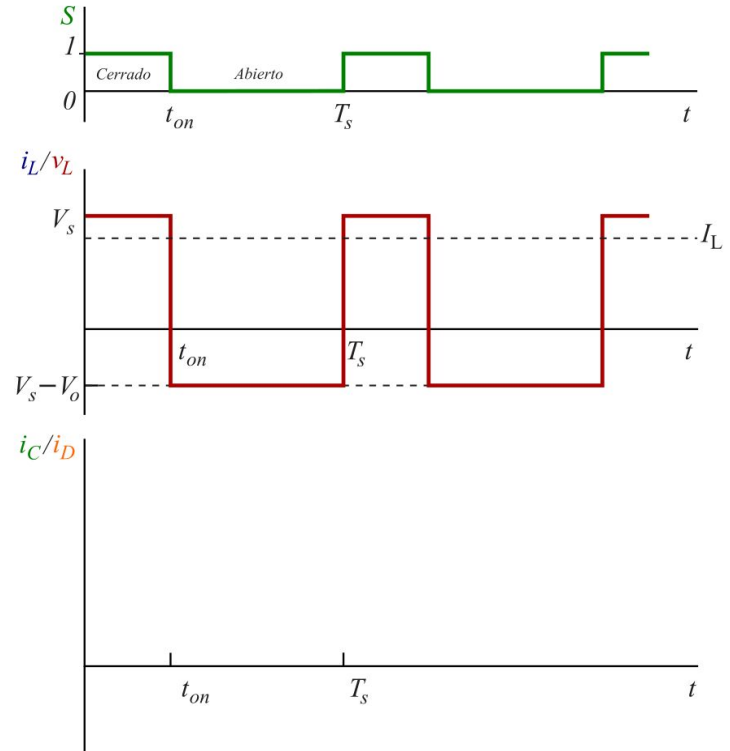
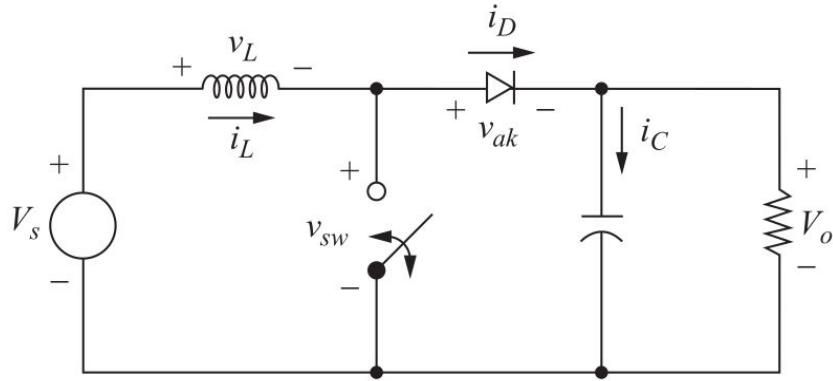
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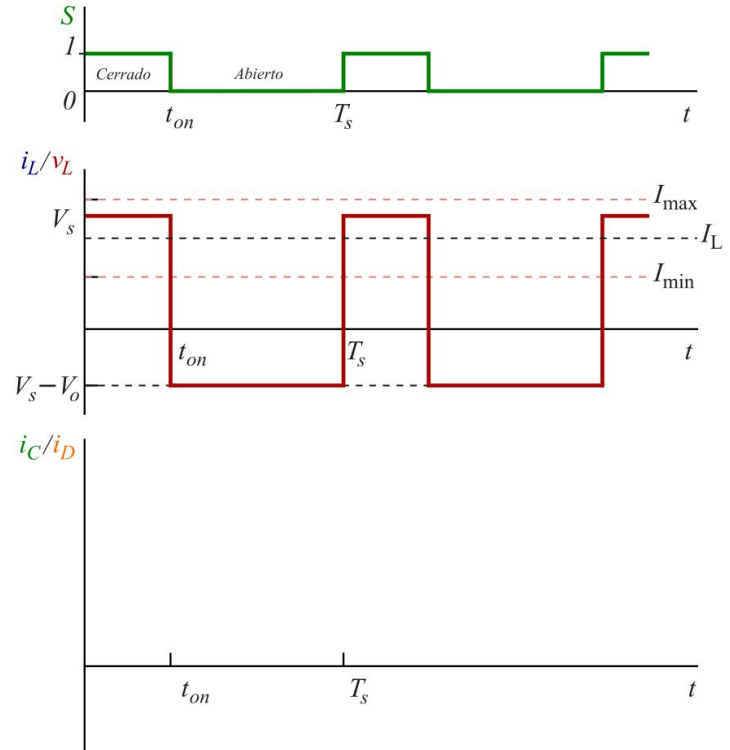
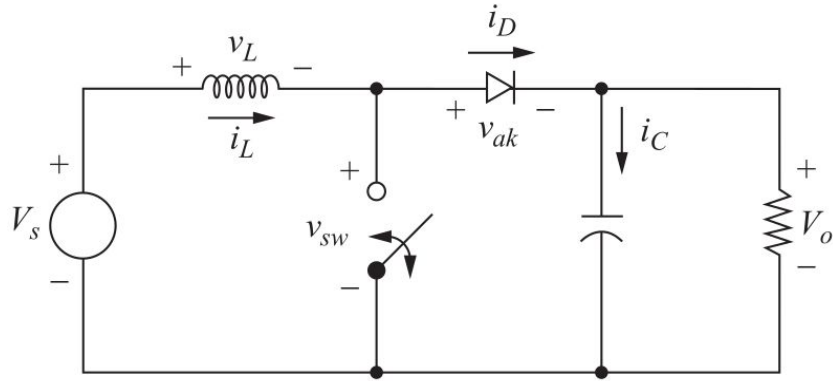
Convertidor Elevador: formas de onda



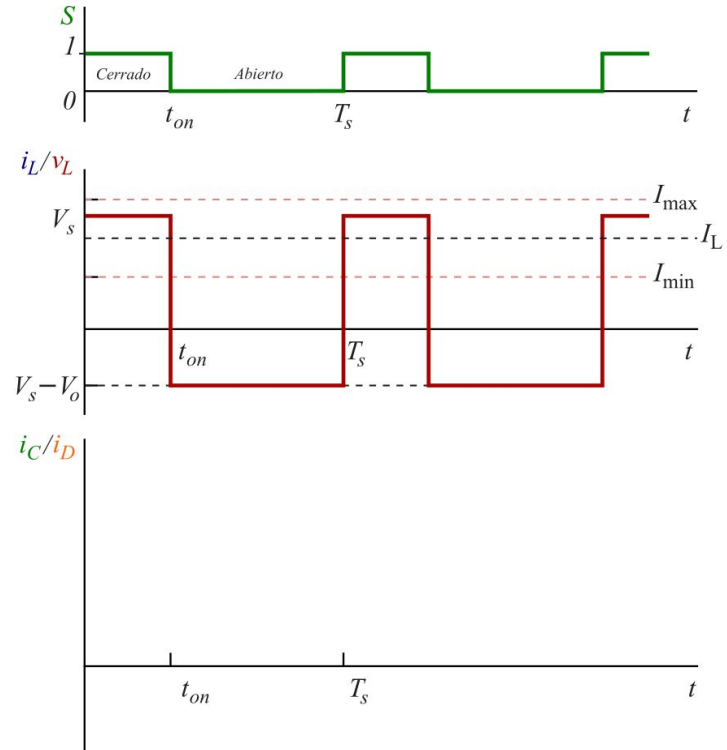
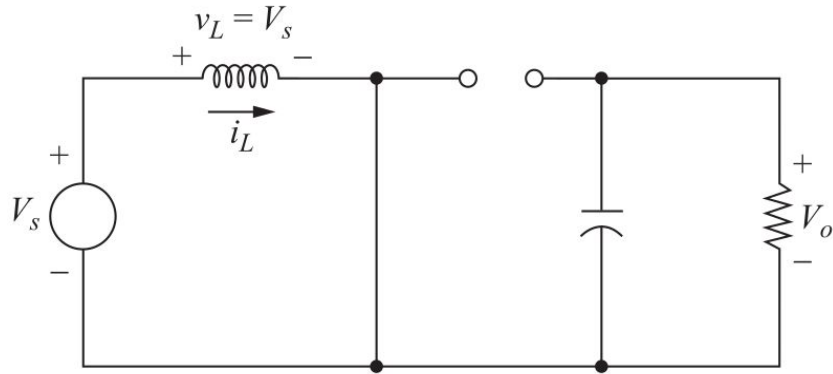
Convertidor Elevador: formas de onda



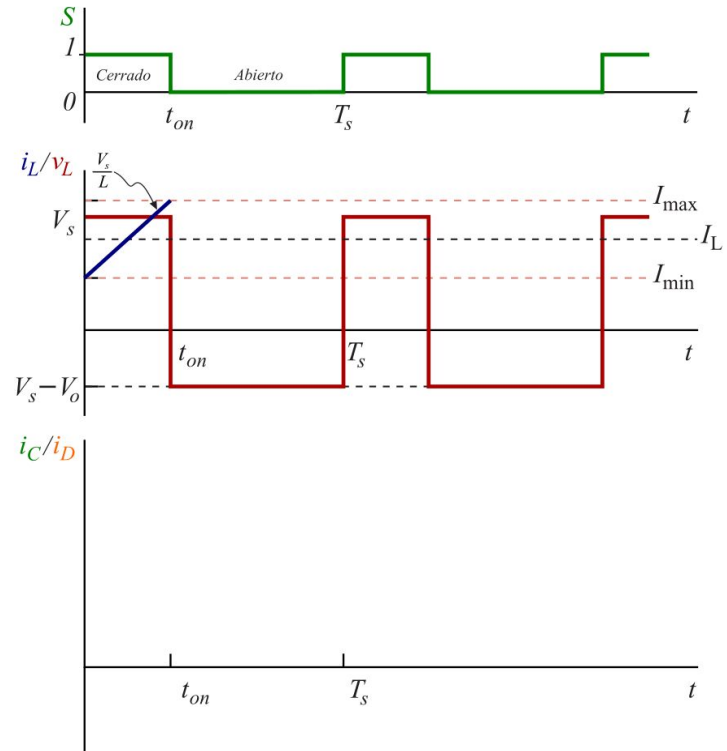
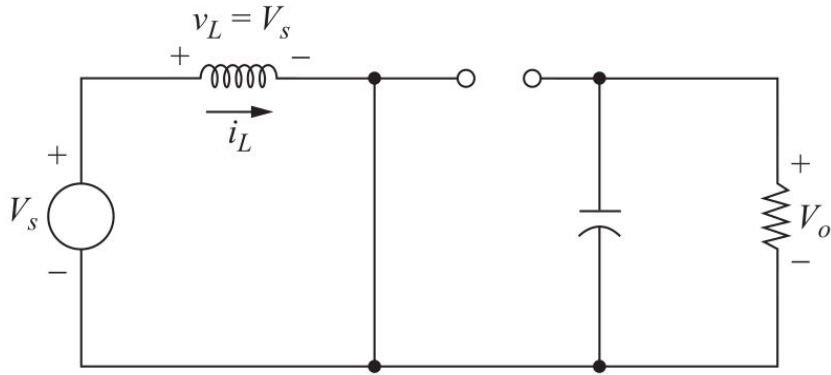
Convertidor Elevador: formas de onda



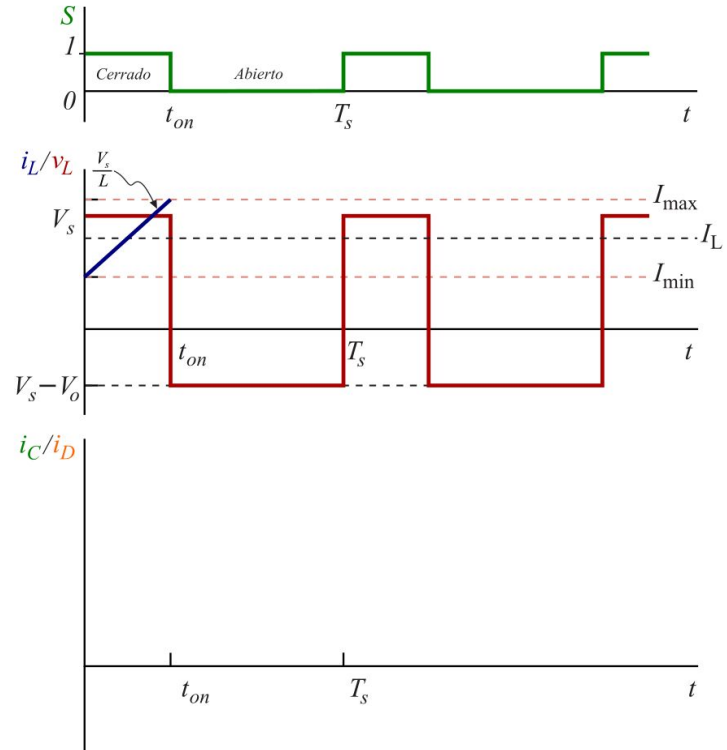
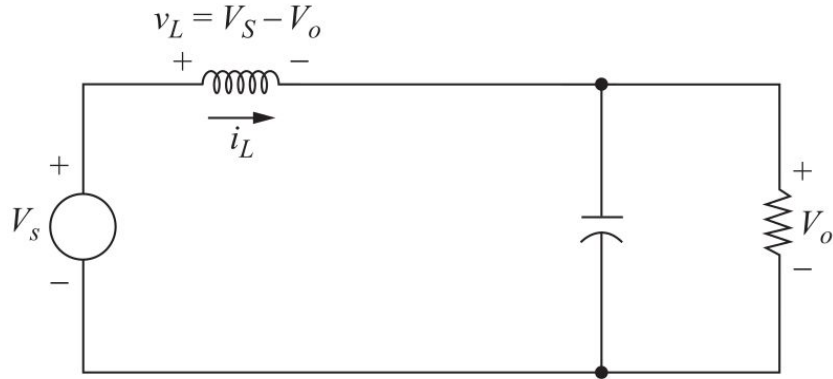
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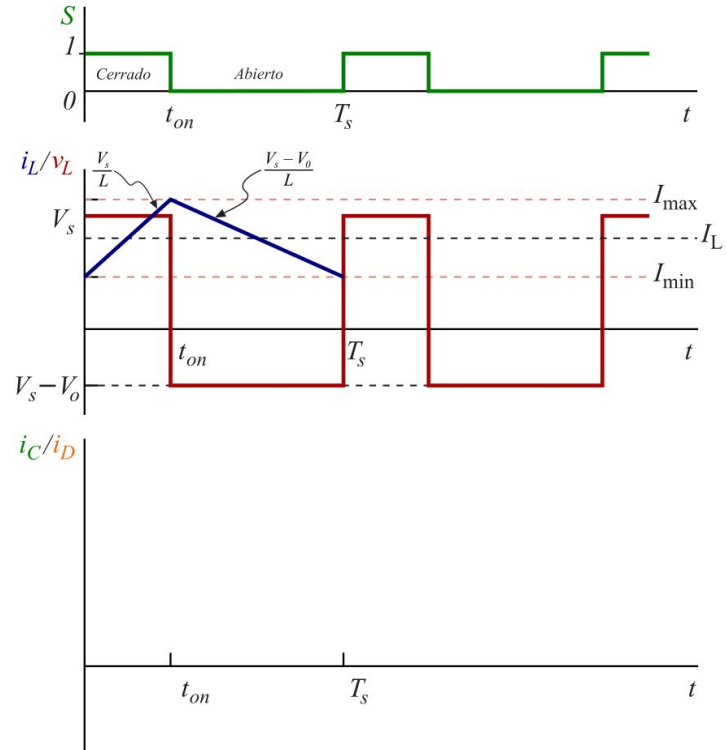
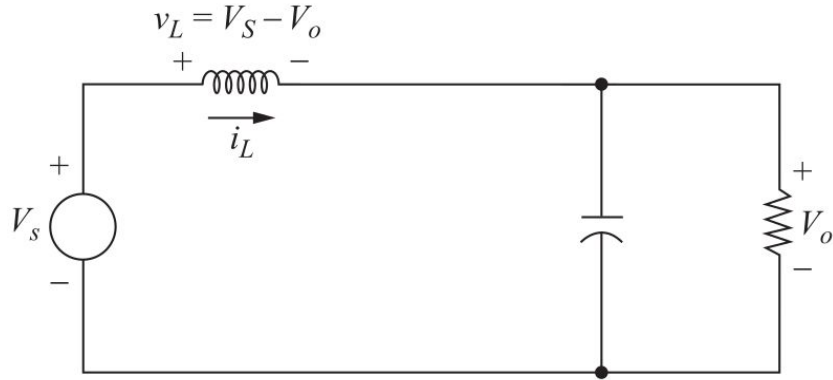
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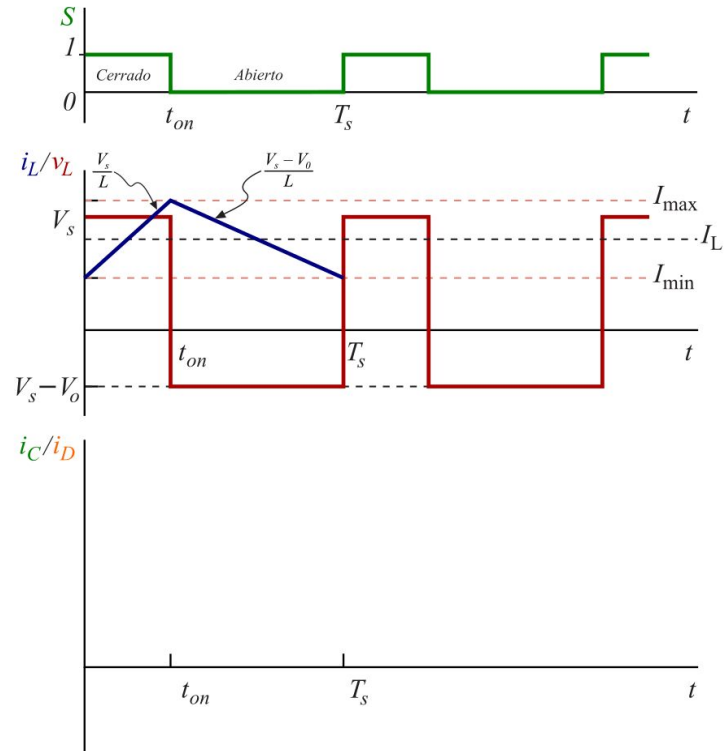
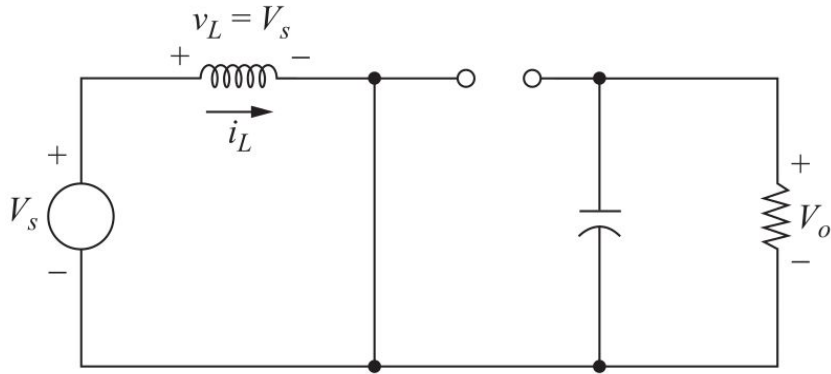
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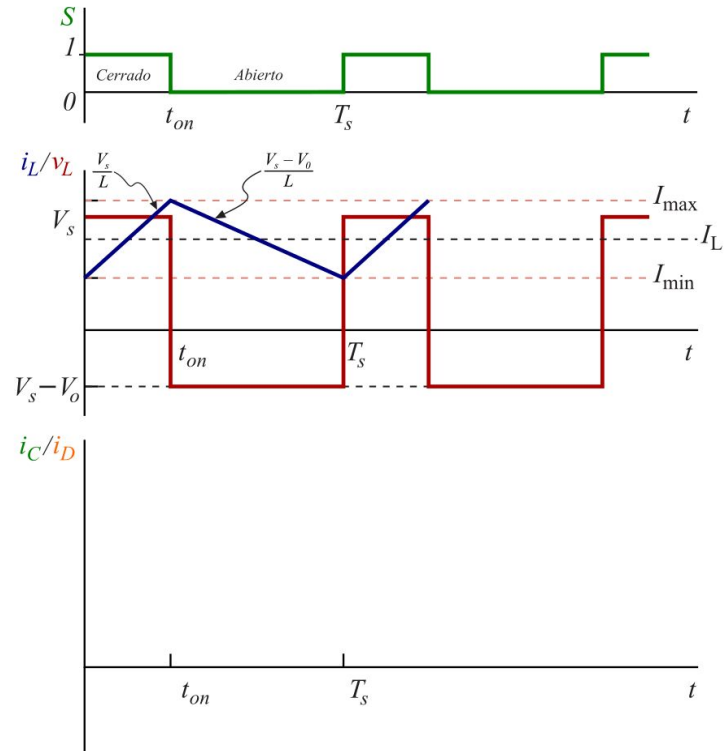
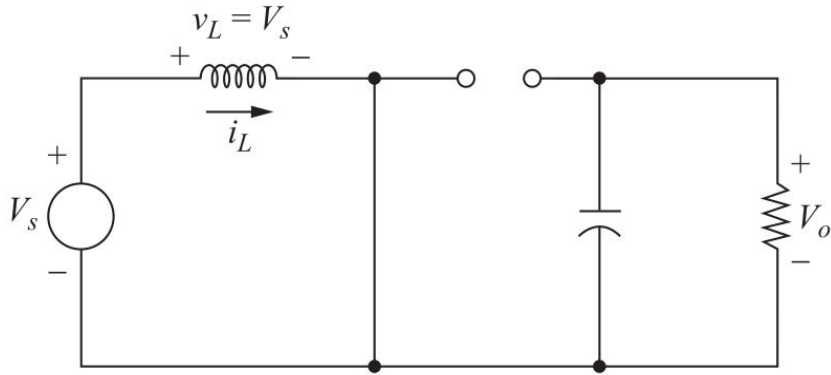
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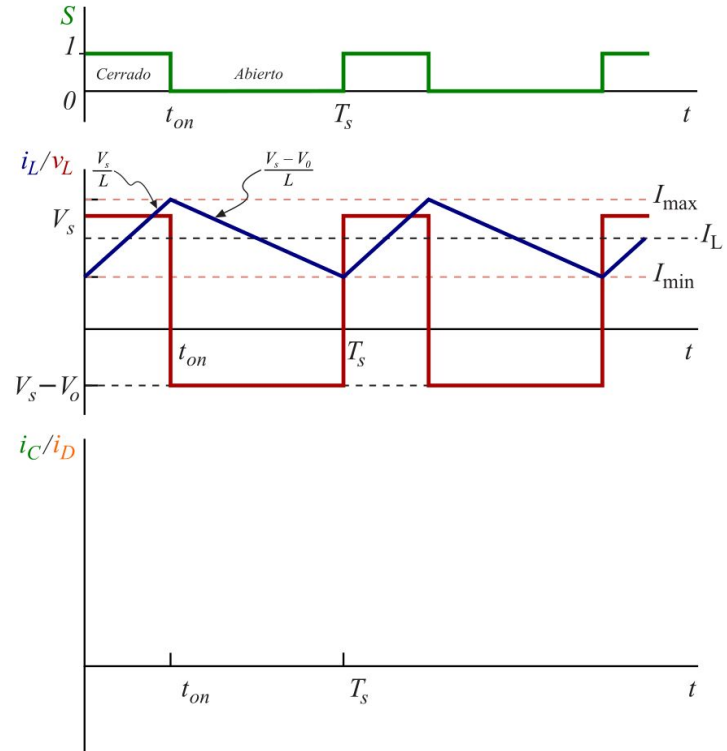
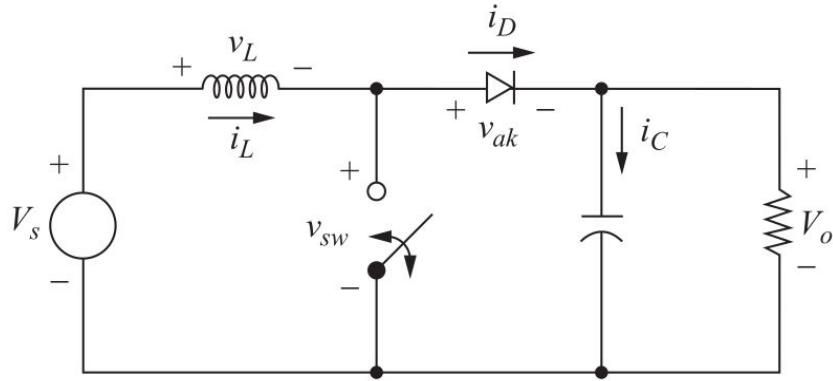
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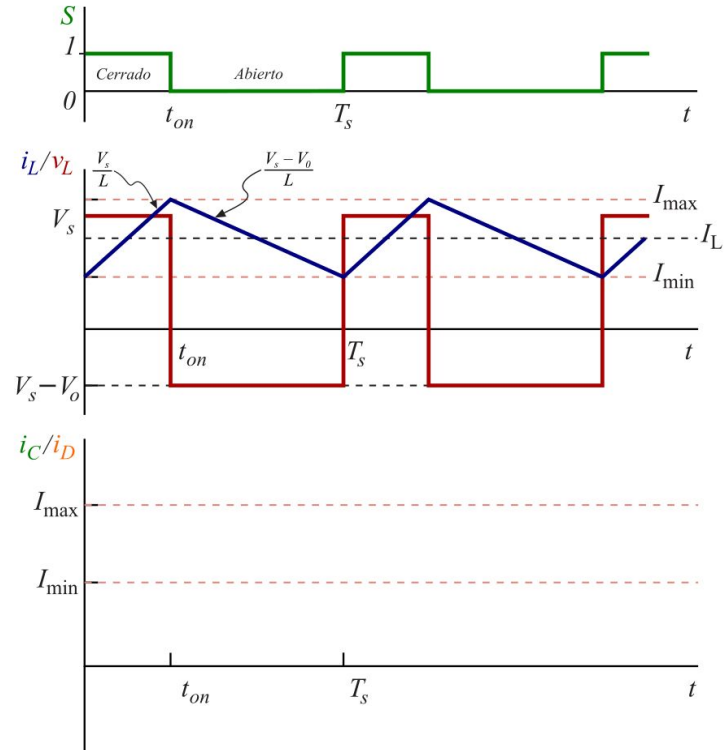
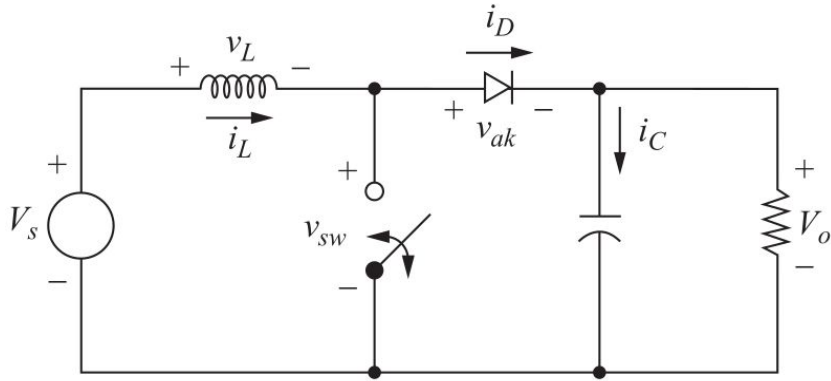
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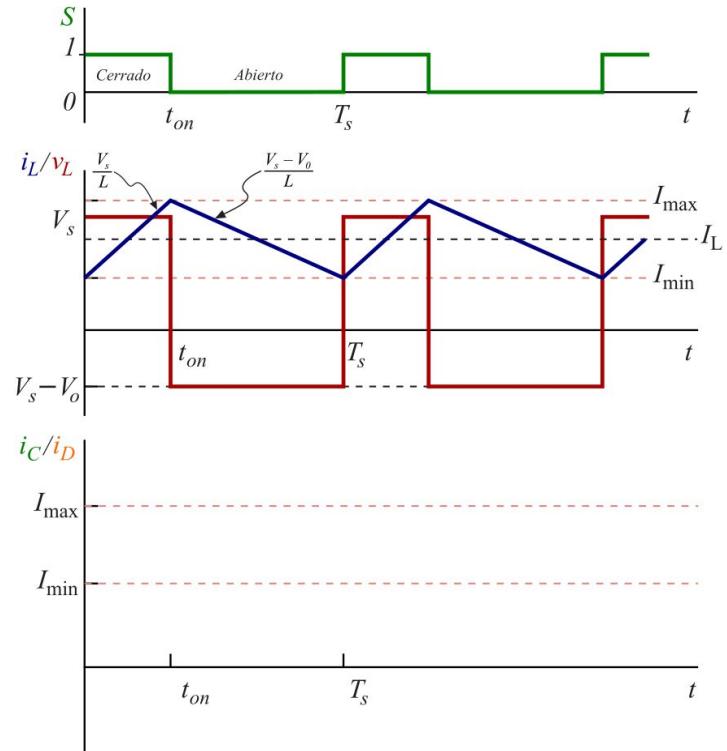
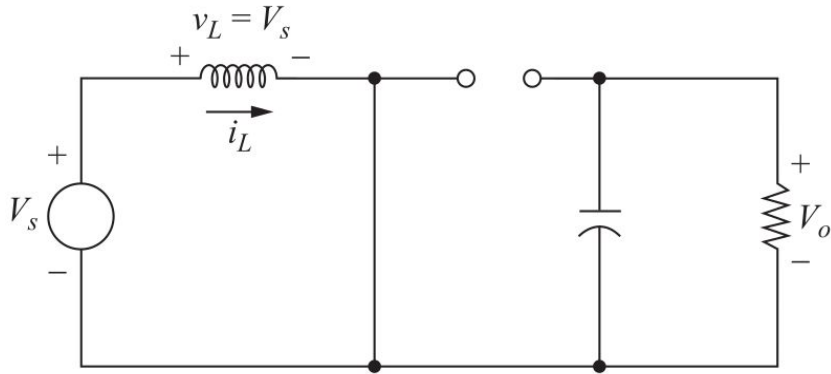
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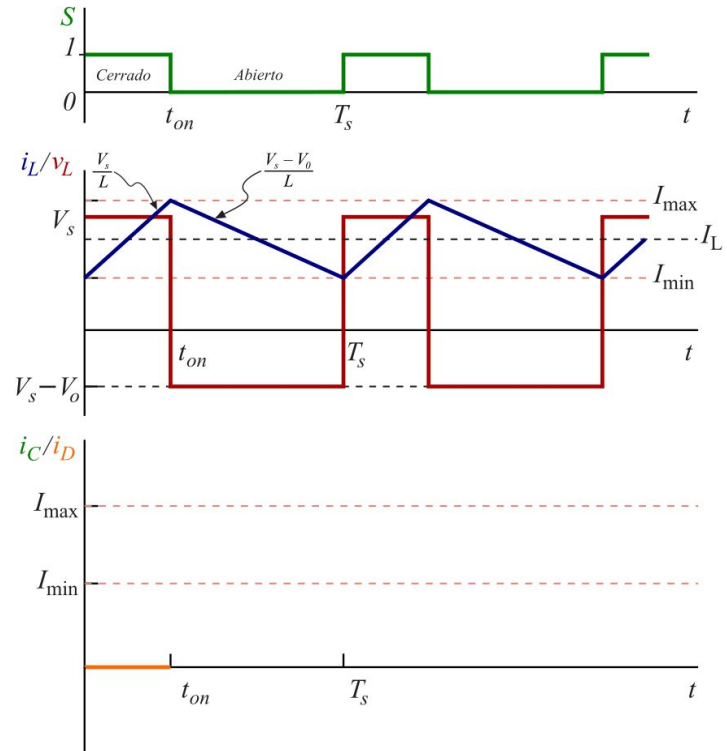
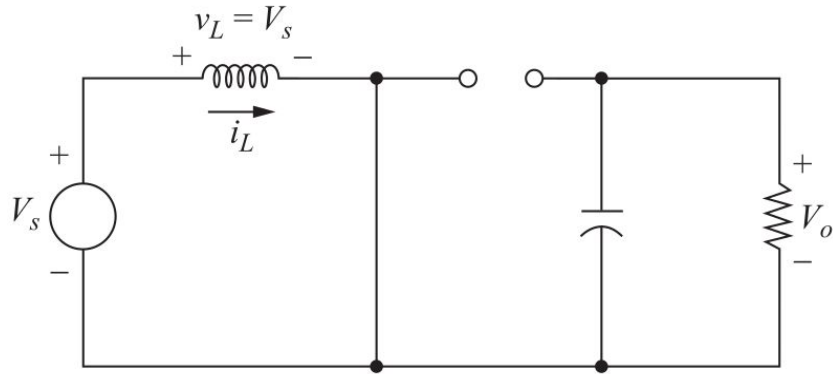
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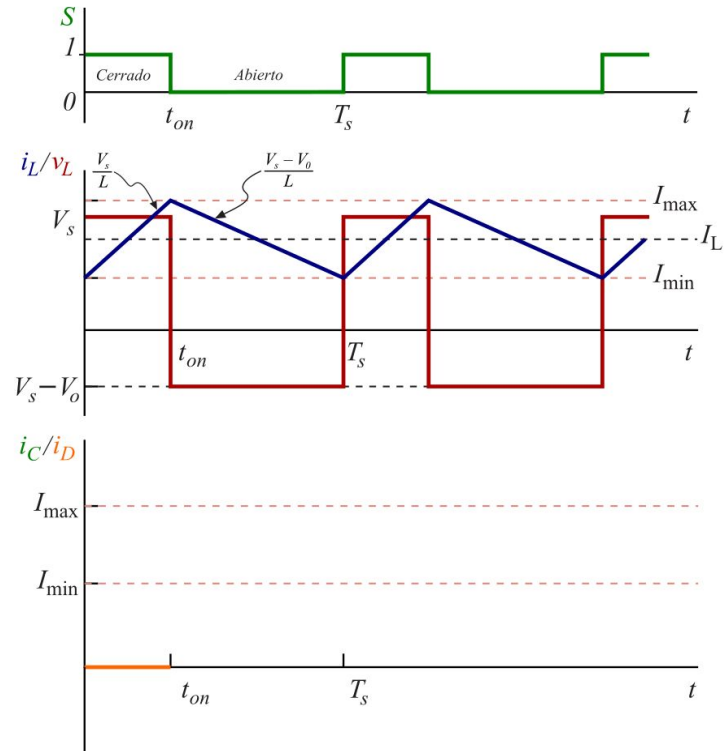
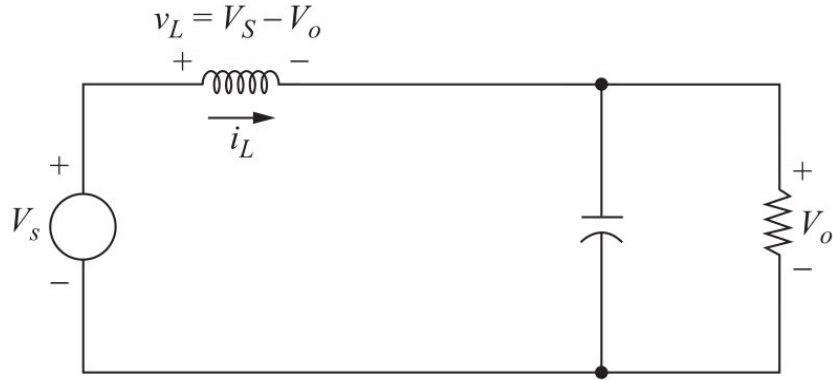
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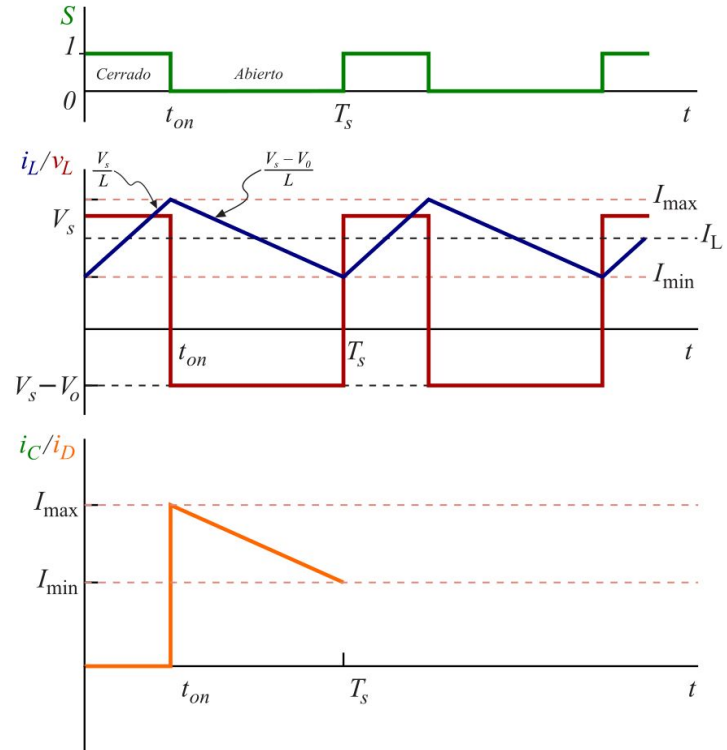
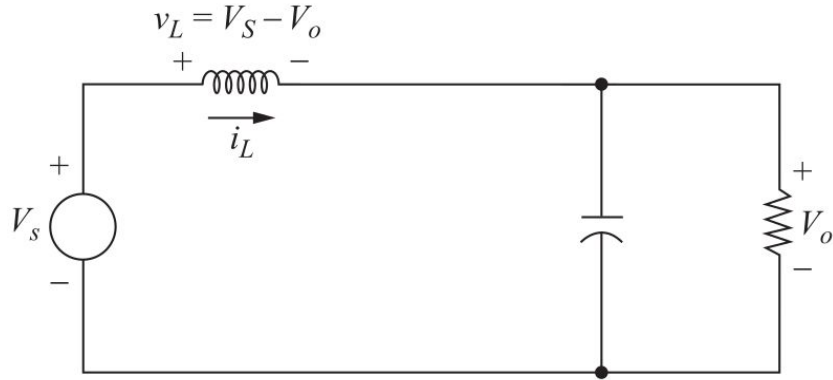
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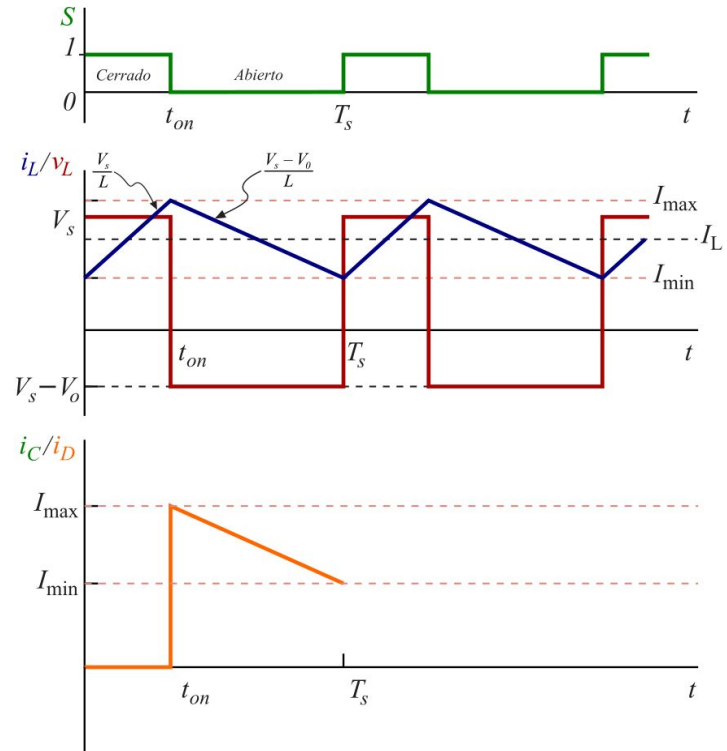
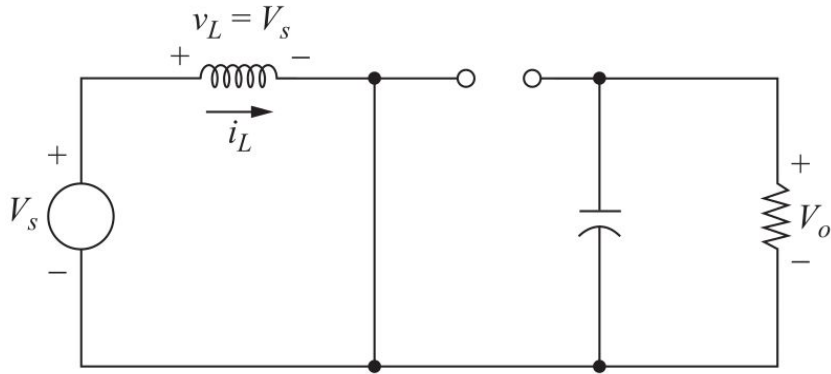
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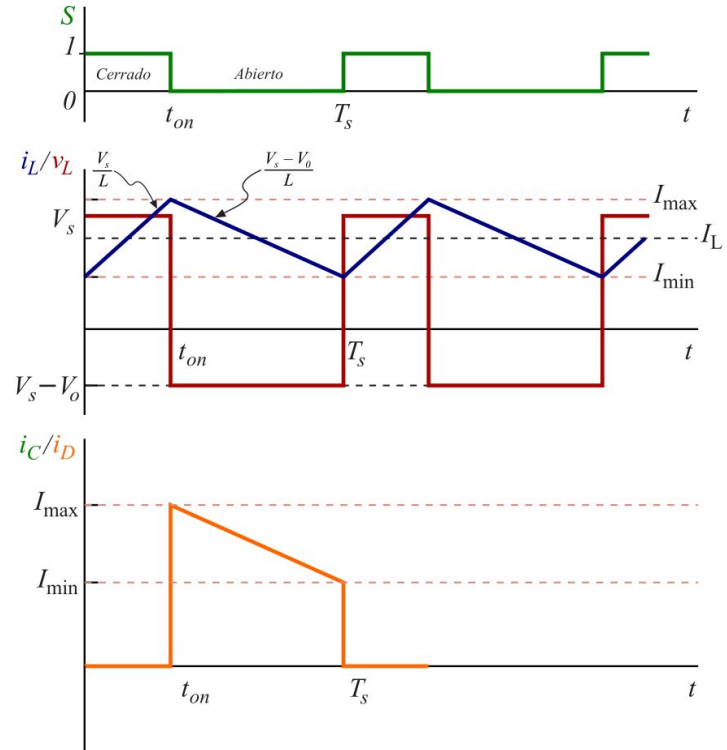
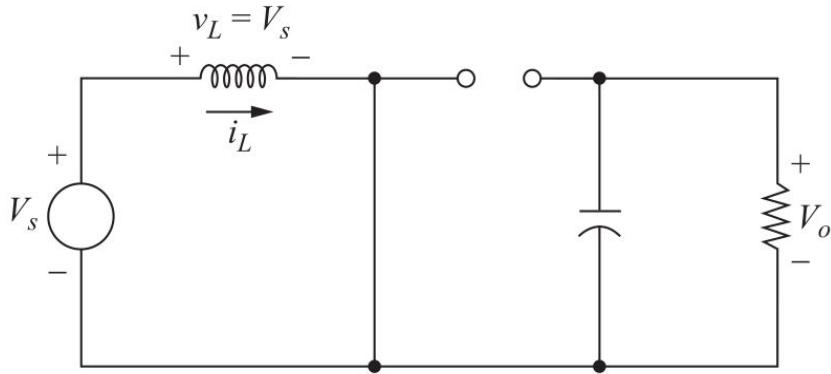
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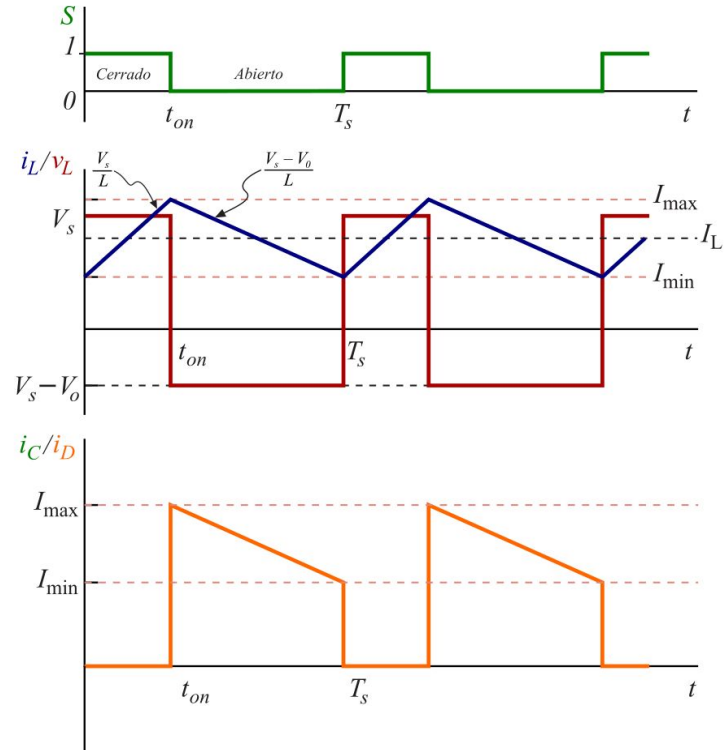
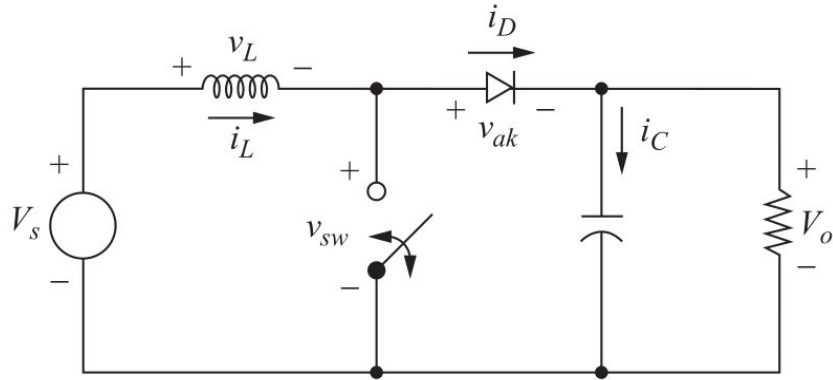
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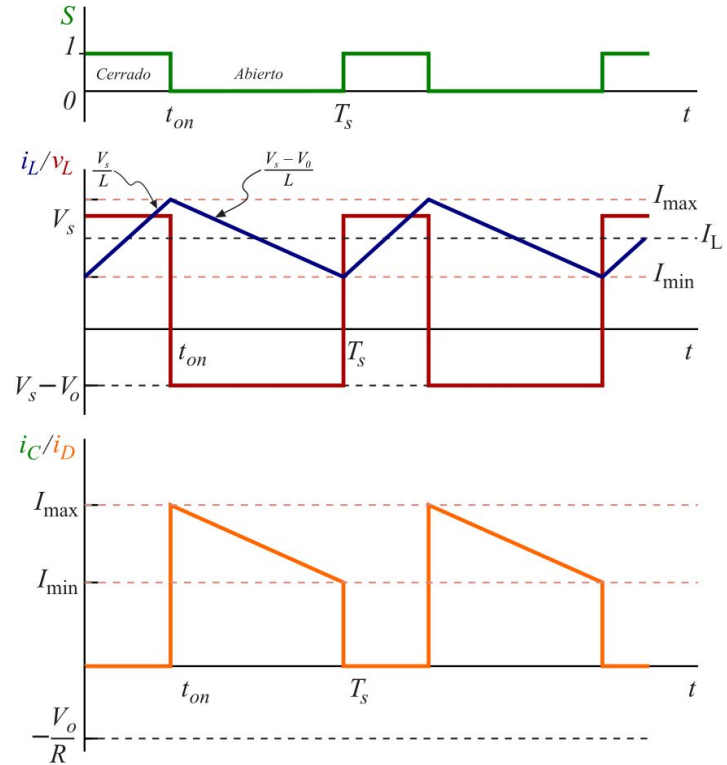
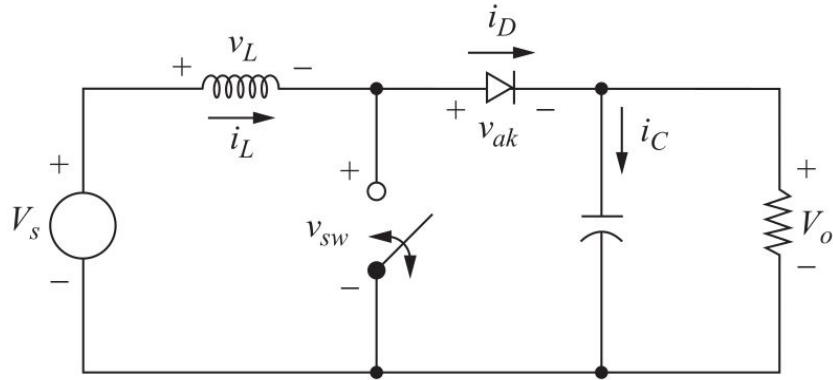
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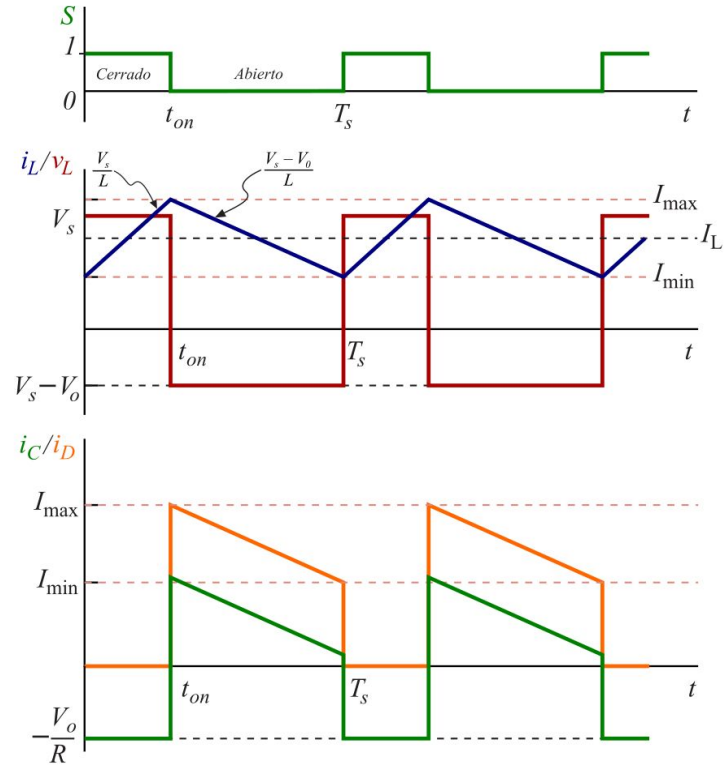
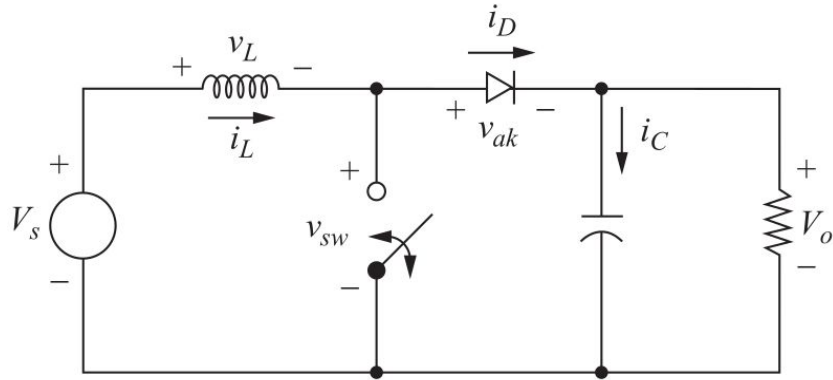
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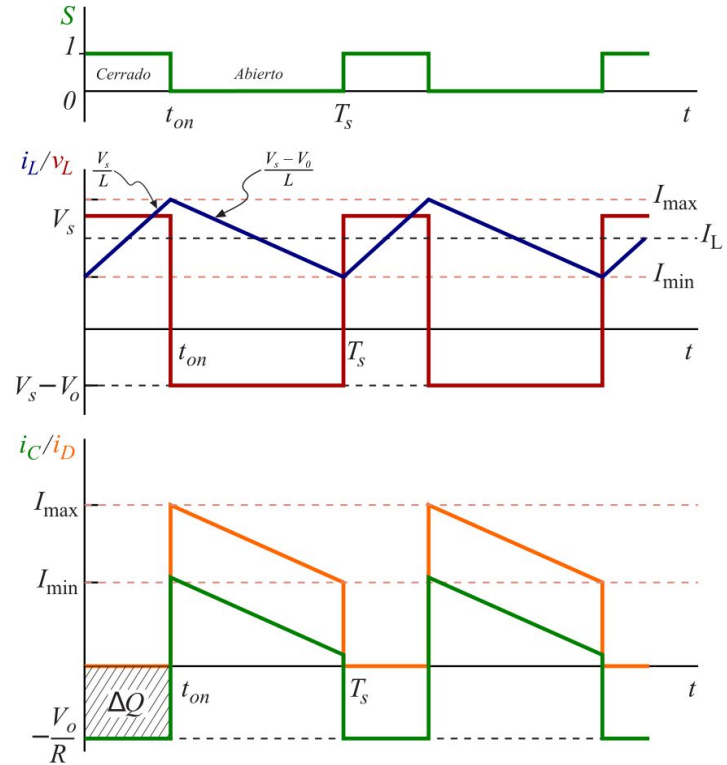
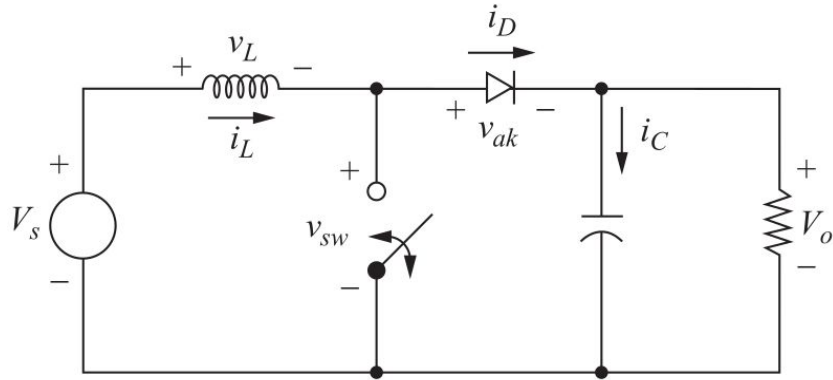
Convertidor Elevador: formas de onda



Convertidor Elevador: formas de onda

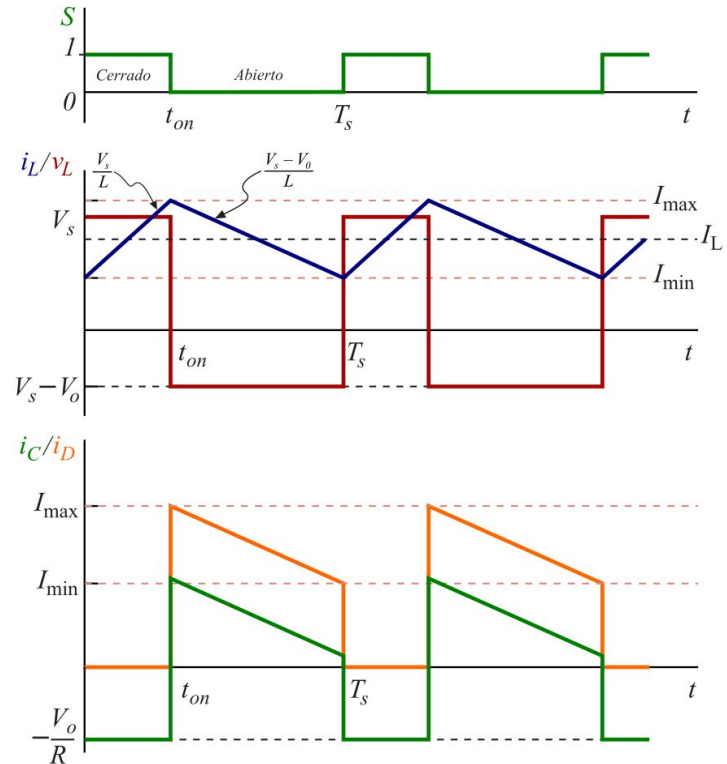


Convertidor Elevador: formas de onda



Convertidor Elevador: relación de conversión

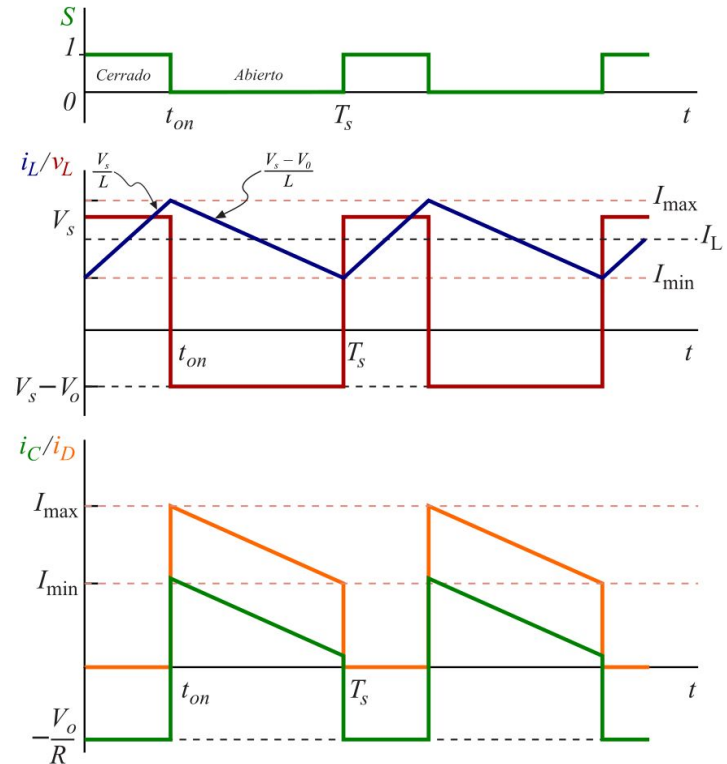
La tensión media en el inductor debe ser cero:



Convertidor Elevador: relación de conversión

La tensión media en el inductor debe ser cero:

$$V_s t_{on} = (V_0 - V_s)(T_s - t_{on})$$

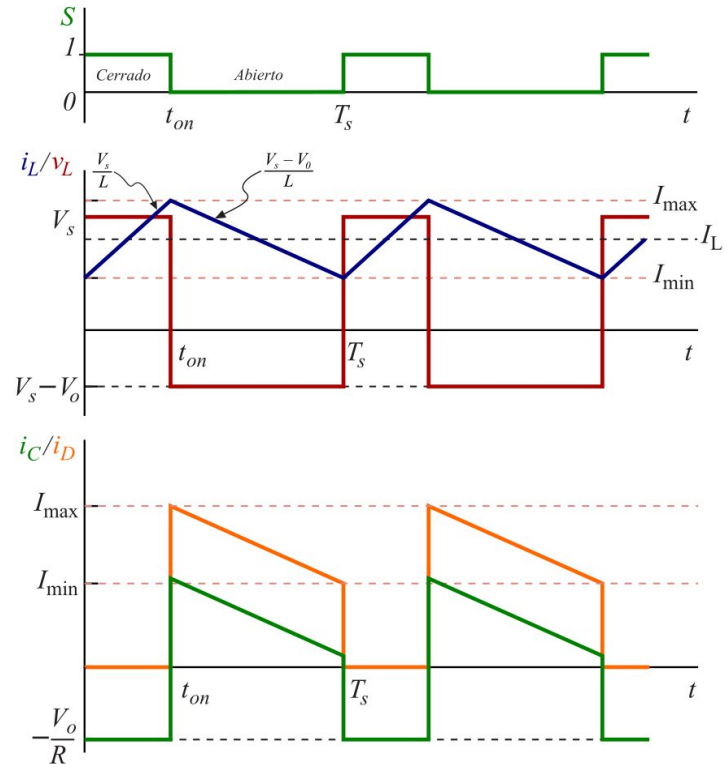


Convertidor Elevador: relación de conversión

La tensión media en el inductor debe ser cero:

$$V_s t_{on} = (V_0 - V_s)(T_s - t_{on})$$

$$V_s t_{on} = V_0 T_s - V_0 t_{on} - V_s T_s + V_s t_{on}$$

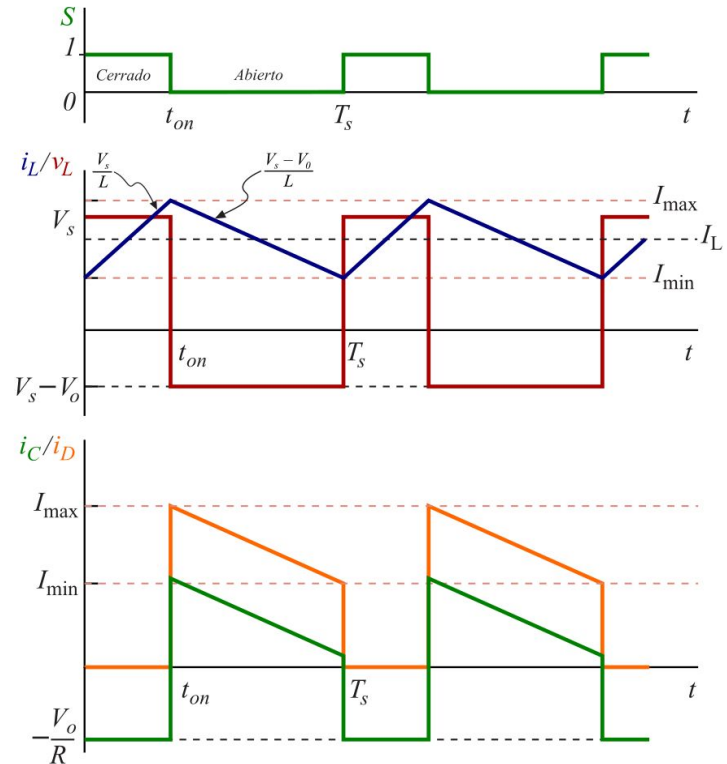


Convertidor Elevador: relación de conversión

La tensión media en el inductor debe ser cero:

$$V_s t_{on} = (V_0 - V_s)(T_s - t_{on})$$

$$\cancel{V_s t_{on}} = V_0 T_s - V_0 t_{on} - V_s T_s + \cancel{V_s t_{on}}$$



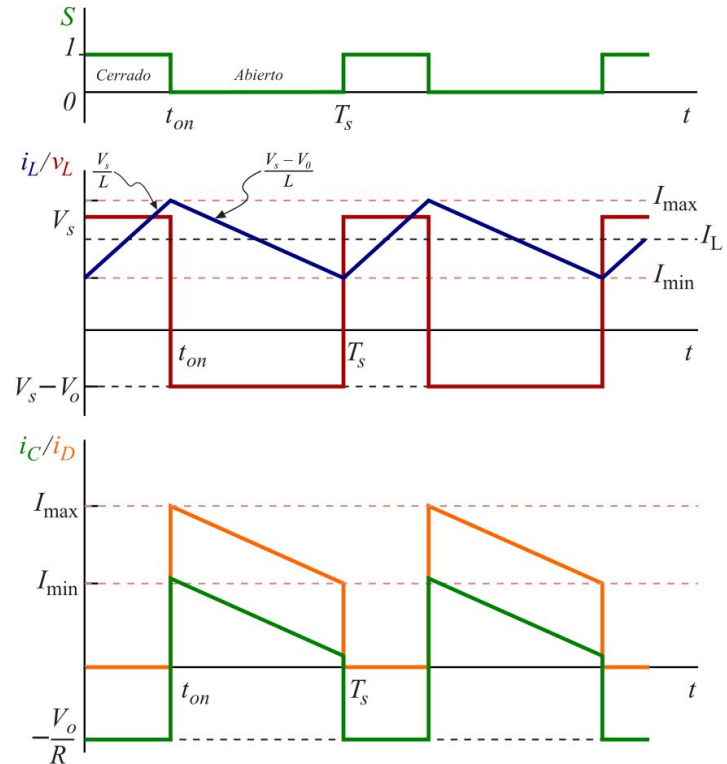
Convertidor Elevador: relación de conversión

La tensión media en el inductor debe ser cero:

$$V_s t_{on} = (V_0 - V_s)(T_s - t_{on})$$

$$\cancel{V_s t_{on}} = V_0 T_s - V_0 t_{on} - V_s T_s + \cancel{V_s t_{on}}$$

$$V_0(T_s - t_{on}) = V_s T_s$$



Convertidor Elevador: relación de conversión

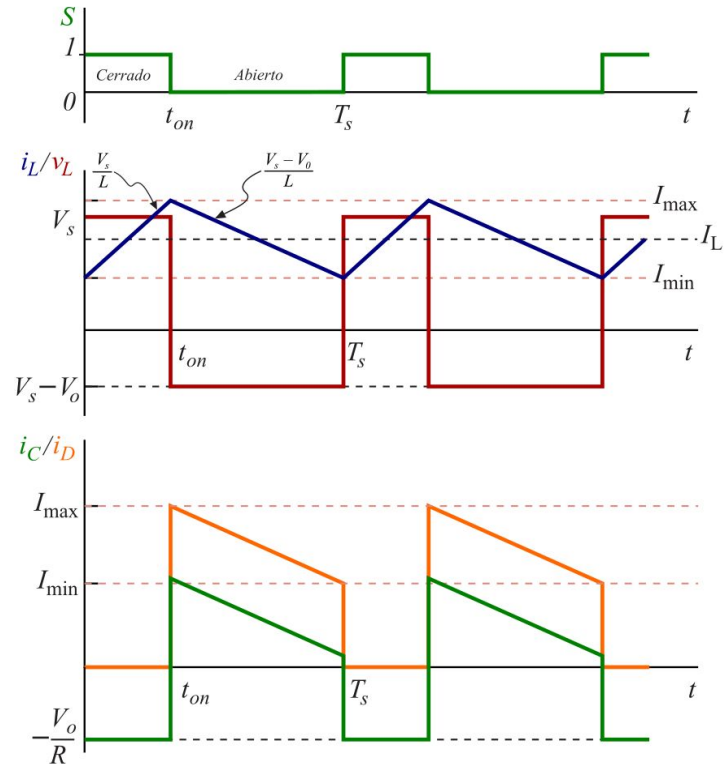
La tensión media en el inductor debe ser cero:

$$V_s t_{on} = (V_0 - V_s)(T_s - t_{on})$$

$$\cancel{V_s t_{on}} = V_0 T_s - V_0 t_{on} - V_s T_s + \cancel{V_s t_{on}}$$

$$V_0(T_s - t_{on}) = V_s T_s$$

$$V_0 \left(1 - \frac{t_{on}}{T_s}\right) = V_s$$



Convertidor Elevador: relación de conversión

La tensión media en el inductor debe ser cero:

$$V_s t_{on} = (V_0 - V_s)(T_s - t_{on})$$

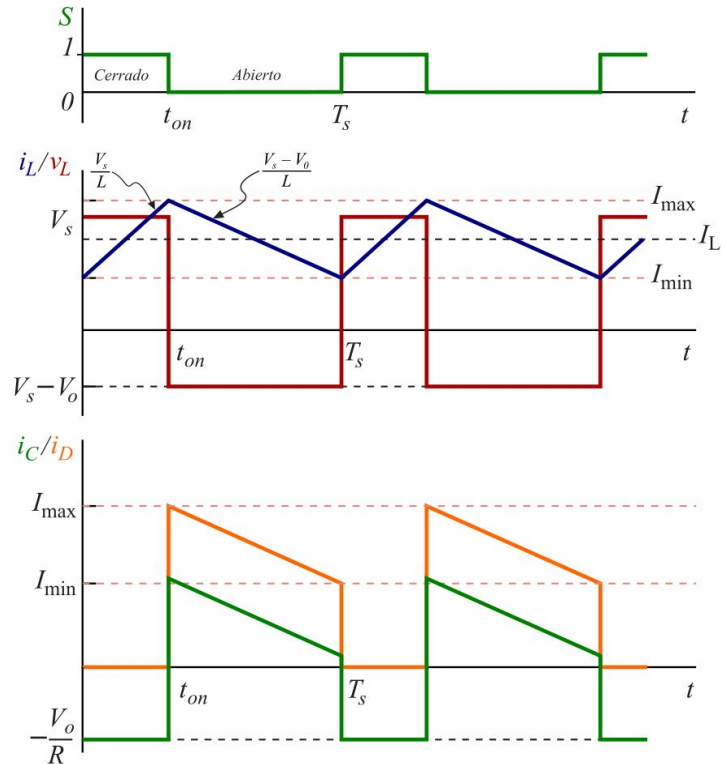
$$\cancel{V_s t_{on}} = V_0 T_s - V_0 t_{on} - V_s T_s + \cancel{V_s t_{on}}$$

$$V_0(T_s - t_{on}) = V_s T_s$$

$$V_0 \left(1 - \frac{t_{on}}{T_s}\right) = V_s$$

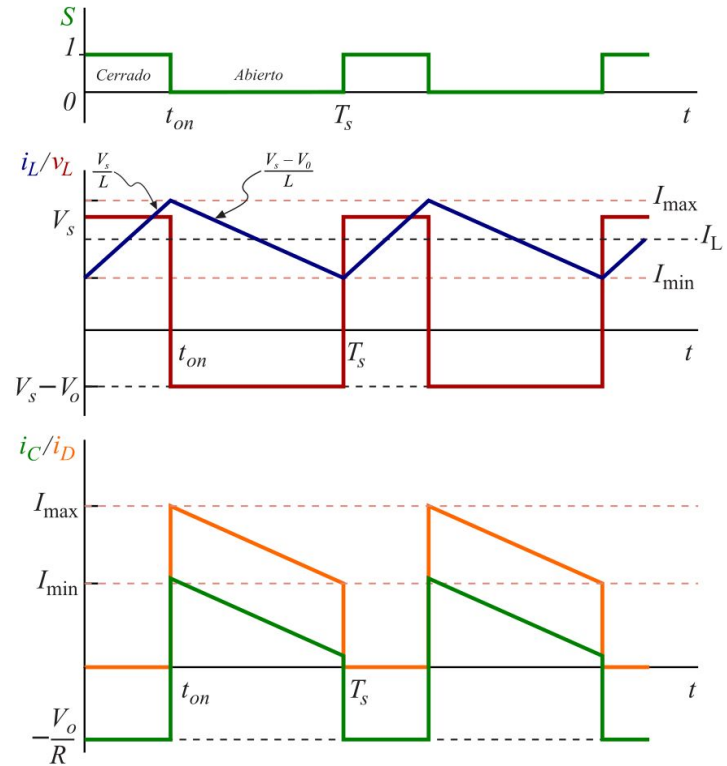
$$V_0(1 - D) = V_s$$

$$V_0 = \frac{V_s}{(1 - D)}$$



Convertidor Elevador: relación de conversión

¿Qué pasa con la relación de corrientes?



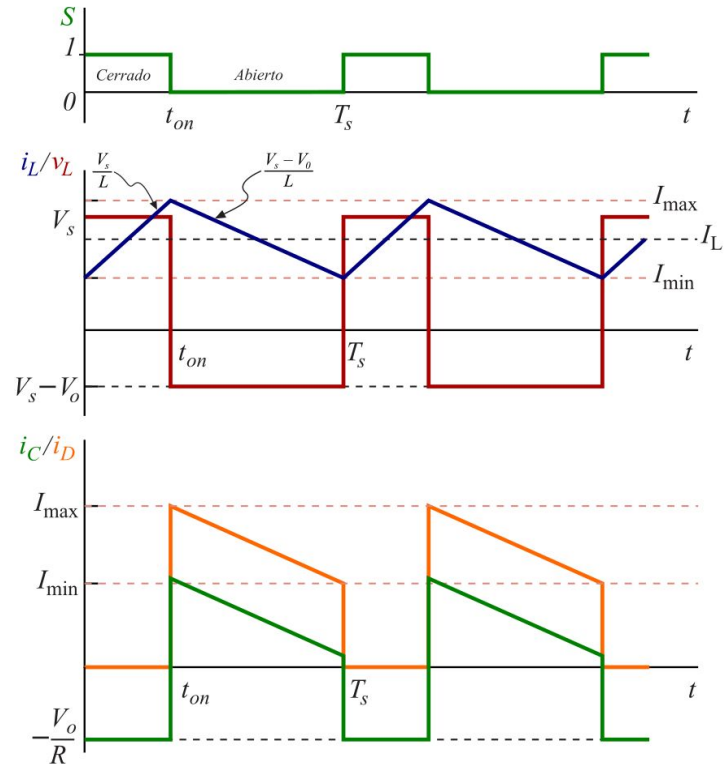
Convertidor Elevador: relación de conversión

¿Qué pasa con la relación de corrientes?

$$P_s = P_o$$

$$V_s I_s = V_o I_o$$

$$\frac{I_s}{I_o} = \frac{V_o}{V_s}$$



Convertidor Elevador: relación de conversión

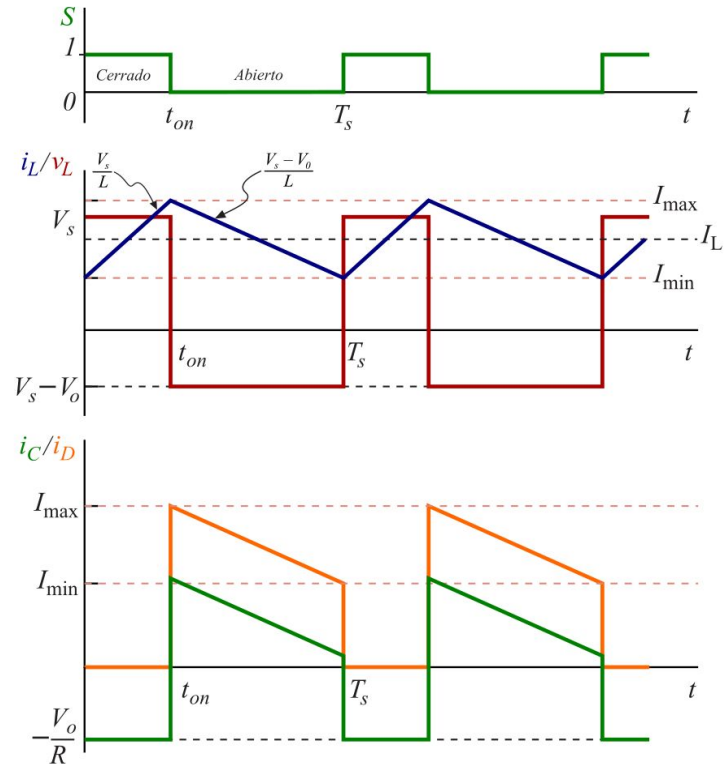
¿Qué pasa con la relación de corrientes?

$$P_s = P_o$$

$$V_s I_s = V_o I_o$$

$$\frac{I_s}{I_o} = \frac{V_o}{V_s}$$

$$V_o = \frac{V_s}{(1 - D)}$$



Convertidor Elevador: relación de conversión

¿Qué pasa con la relación de corrientes?

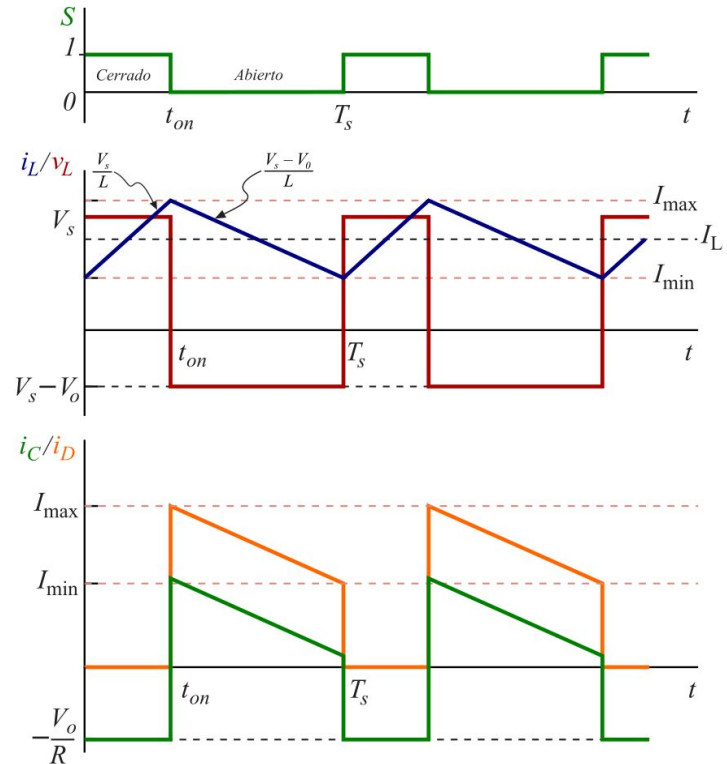
$$P_s = P_o$$

$$V_s I_s = V_o I_o$$

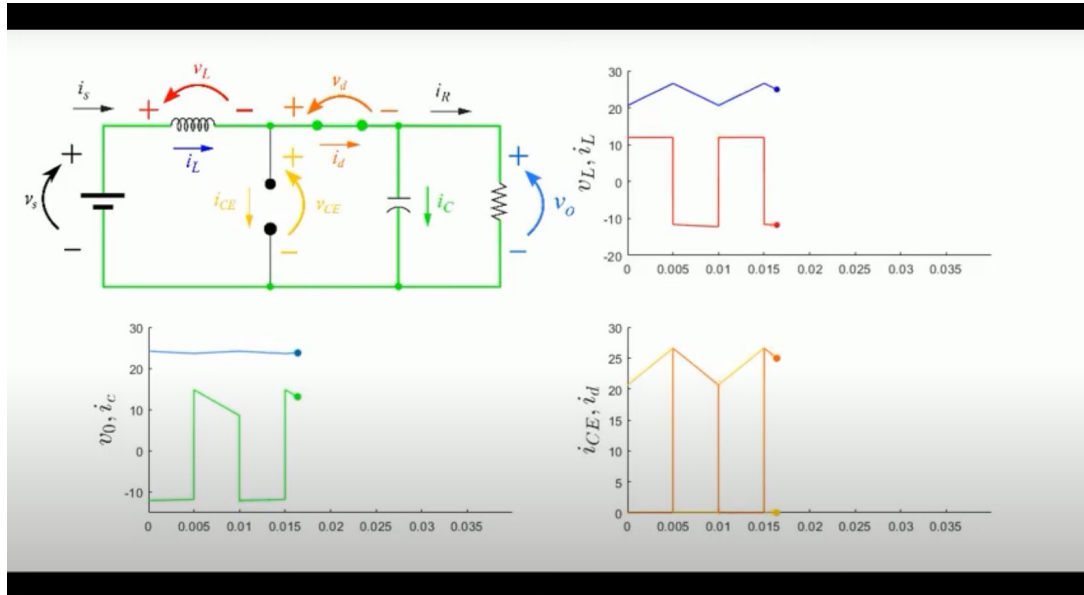
$$\frac{I_s}{I_o} = \frac{V_o}{V_s}$$

$$V_o = \frac{V_s}{(1 - D)}$$

$$I_s = \frac{I_o}{(1 - D)}$$



Convertidor Elevador: formas de onda



<https://youtu.be/ef62il9J17U>



Bibliografía

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